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EduVerse

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Abstract

The primary goal of EduVerse is to combine the best features of renowned e-learning platforms such as Coursera, Udemy, Hackerrank, Nearpeer, and Educative into a single platform with the aim of providing learners with an all-inclusive educational solution. Our web-application based platform provides custom learning paths for every individual based on their weaknesses, strengths and interests. We will be focusing on providing an all-inclusive evaluation of our students for companies looking to recruit, which includes giving detailed performance breakdowns. Moreover, we will be having an integrated coding environment available for the students to complete assignments and practice their coding, also encouraging some healthy competition based learning.

Executive Summary

EduVerse is a web application based online learning platform designed to provide a personalized educational experience. This project provides personalized learning paths, skill assessment, code editing environments, competitive learning, and tools for recruitment and skill evaluation. The system helps both instructors and learners by offering features such as course creation, progress monitoring, assessments, and analytics. Users can create accounts, scroll through our courses, enroll, complete assignments, and access a ton of learning resources.

Our project is an innovative online learning platform which we have designed to change the way online learning works. We plan on doing this by addressing the issues that are in this field. Firstly, we deal with the issue of not having personalized paths for everyone based on their own skill set, ensuring that their paths align with their interests. Secondly, we help with the recruitment process by providing you and the companies with a detailed performance breakdown. This breakdown helps recruiters to assess the students on their merits. Moreover, EduVerse helps students to transform their knowledge into skills with the help of an integrated code editor. Lastly, EduVerse addresses the issue of learner engagement by motivating students into competition based learning, creating an interactive and healthy environment to enhance learning outcomes. In short, EduVerse plans to transform online learning by improving existing ways by making learning more personalized, practical, and engaging.

EduVerse presents a solution to the existing challenges in both traditional and online education. Firstly, EduVerse recognizes the limitation of traditional education models and offers a solution through Personalized Learning Paths. We ensure that learning is more beneficial to every student by making their paths, keeping their interests and skills in mind. Secondly, EduVerse addresses the lack of good assessment methods by providing a precise evaluation of an individual's skills and abilities. Furthermore, our platform aims to help learners transform their knowledge into practical skills by providing them with an integrated coding environment.

Related work has also been covered in one of our chapters. It discusses strengths, weaknesses, and how we relate to already established online learning platforms. We started with Coursera, focusing on its vast collection of courses but it surely is a little expensive if you want a certificate. Next, we discussed Educative's hand on learning approach with technical courses only being offered. Neerpeer's community discussion forums are to be appreciated, whereas the courses they offer are completely different from EduVerse catalog. HackerRank is recognized for its competition-based learning, but it's noted that some challenges may be tough for beginners. Moreover, Codecademy is appreciated for its interactive coding exercises whereas Khan Academy's commitment to providing free, high-quality education is great. This

Chapter basically explains why EduVerse's goal of personalized learning is so important. Lastly, This section serves as a valuable example to understand how EduVerse aims to positions itself among its competitors.

The system's functional requirements include user authentication, content management, personalization, assessment, and communication. And non-functional requirements include responsive performance, high uptime, user-friendly interfaces, and modular code design. Some assumptions we make are that users have JavaScript-compatible browsers and internet access. Hardware and software requirements include a desktop or laptop with an internet connection and specific software tools. Detailed use cases cover various user interactions, from registration and login to course enrollment, live session scheduling, and feedback provision. Additionally, the document addresses account settings management, certificate issuance, quiz competitions, and resource recommendations.

Furthermore, the database design is outlined through an Entity-Relationship (ER) diagram and a data dictionary, presenting the schema for entities like instructors, learners, and courses, along with their attributes and relationships. The Graphical User Interface (GUI) section provides visual representations of the platform's user interactions, making it easier to comprehend the user experience flow. Lastly, risk analysis identifies potential technical, business, content and quality, user adoption, and competition risks. To mitigate these risks effectively, the document recommends continuous risk assessments, response plans, adaptability, and flexibility throughout the project's lifecycle. EduVerse aims to redefine online learning, offering a dynamic and personalized educational experience to learners, instructors, and organizations worldwide.

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Chapter 1 Introduction

There are many e-learning platforms that provide opportunities for learners to learn new skills or polish their existing skills. With the COVID-19 pandemic in 2020, the whole world shifted online and learners found themselves surrounded by different e-learning platforms that provided different learning experience compared to traditional face-to-face learning.
Our website will be an all-inclusive learning platform that will combine the likes of different learning platforms. In addition to this, the learning platform will recommend students about their current academic performance and recommend different courses depending on their preferences and their strengths and weaknesses.

1.1 Purpose of this Document

This document intends to elaborate the endeavors of ”EduVerse,” an innovative and ambitious learning platform designed to revolutionize the realm of online education. This documentation discusses in detail the concept, design, development, and potential impact of EduVerse. Its purpose is to provide a clear and structured framework for understanding the project's scope, goals, and significance. The project aims to merge the best features of different e-learning platforms such as Coursera, Udemy, Hackerrank, Nearpeer, and Educative into a single platform with the aim of providing learners with an all-inclusive educational solution.

1.2 Intended Audience

The intended audience for this document encompasses a diverse group of stakeholders who will contribute to, evaluate, or benefit from the development and implementation of EduVerse.

1.2.1 Academic and Educational Communities

- Educators and Instructors: The goal of EduVerse is to give teachers an efficient tool for creating a successful learning experience. This document will help educators understand how EduVerse can improve their teaching methods.
- Students: EduVerse is primarily developed to cater to the educational needs of students. This documentation will help students understand how EduVerse can help them in achieving their academic goals more effectively.

1.2.2 Technology Enthusiasts

- Developers: Software developers will find this document valuable for understanding the technical aspects of EduVerse, including the integrated code editor and the underlying technology stack.
- Innovators and Entrepreneurs: Individuals and organizations seeking opportunities in the educational technology sector will gain insights into the market potential and innovative features of EduVerse.

1.2.3 Employers and Recruiters

- Human Resource Professionals: HR personnel looking for innovative ways to assess and recruit talent will learn about EduVerse's holistic candidate evaluation system and its potential to identify well-rounded candidates.

1.3 Definitions, Acronyms, and Abbreviations

List all important definitions, acronyms, and abbreviations used in this document.

SDG: Sustainable Development Goal

FYP: Final Year Project

MVP: Minimum Viable Product

GUI: Graphical User Interface

UI: User Interface

UX: User Experience

COVID-19: Corona Virus Disease 2019

HR: Human Resource

IDE: Integrated Development Environment

JS: Javascript

DB: Database

ERD: Entity Relationship Diagram **LMS**: Learning Management System

1.4 Conclusion

The first chapter of this research introduces the concept of "EduVerse" as an all inclusive learning platform, its intended audience and the definitions and abbreviations of complex terms. The second chapter dives deeper into the the vision of the project by discussing the need of an all-in-one learning platform, the goals and objectives of EduVerse. The project scope and sustainable development goals are also

discussed in this chapter. Moreover, the constraints of this project and the business opportunities are also key points in this chapter. Lastly, we have the stakeholder's description and the high-level goals in this chapter. In-depth analysis of related work and literature review are discussed in chapter three and is summarised in the literature review summary table shared at the end of the chapter. The fourth chapter comprises of all the software requirement specifications, the list of features it has. It also highlights the functional and non-functional requirements along with the quality attributes and assumptions. Moreover, it contains the hardware and software requirements. Most importantly, it consists of User Cases with GUI, the database design which mainly contains ER diagram and data dictionary. In addition to that, this chapter includes the Risk analysis.

Chapter 2 Project Vision

In Chapter 2, we dive deeper into the visionary aspects that underpin the EduVerse project. Here, we delve into the complexities of the education and edtech domains, outlining the pressing challenges that EduVerse endeavors to tackle. Through this chapter, we gain insights into the project's objectives, scope, and the inherent constraints it faces. These insights provide a clear perspective on EduVerse's ambitious mission and its potential to transform the educational landscape.

2.1 Problem Domain Overview

The EduVerse project envisions a dynamic solution to the challenges prevailing within the realm of online education. The problem domain it addresses is multifaceted, reflecting the complexities of modern education and the limitations of existing e-learning platforms.
Online education platforms typically offer a one-size-fits-all approach, failing to recognize that every person has unique strengths, weaknesses and attention span. This lack of personalization results in mediocre learning experiences. Moreover, the absence of practical hands-on approach and meaningful evaluation of learners stops the transition from theory to practicality.

2.2 Problem Statement

The main issue that EduVerse seeks to resolve is the incapacity of current online learning systems to provide individualized learning experiences. This issue has numerous difficulties, such as:

- Lack of Personalization: Learners' skills and their preferred career path may not align with current e-learning methods' inability to adjust to the unique needs of each individual student.
- Minimal hands on practice: Current learning environments focus on providing a stream for the instructor to share everything and this doesnt engage the learners and neither does it give them hands on learning experience.

2.3 Problem Elaboration

This section will provide an understanding of the different sub-problems that the EduVerse project seeks to address as well as a deeper dive into the main problem that was highlighted in the previous section.

2.3.1 Lack of Personalization

Traditional e-learning platforms often adopt a one-size-fits-all approach, offering standardized courses that do not adapt to individual learners' unique needs, strengths, and weaknesses. This lack of personalization can result in a gap between a learner's skills and their chosen career path. For example, a student with an interest for programming may find themselves following a basic curriculum that does not align with their plans. EduVerse seeks to address this sub-problem by tailoring learning paths based on individual strengths and weaknesses.

2.3.2 Ineffective Candidate Assessment

The existing recruitment process heavily relies on resumes and interviews to evaluate candidates. However, these methods often provide an incomplete picture of an applicant. This lack of understanding becomes especially pronounced in fields where skills are important, such as software development or engineering. EduVerse understands the need for more accurate performance breakdowns, helping recruiters to gain detailed insights into an applicant's abilities.

2.3.3 Engagement and Transition to Practical Skills

Another challenge is eliminating the gap between knowledge gained through online courses and the practical skills required in the industry. Many online learning platforms mainly focus on theory which leaves learners unprepared for reality. EduVerse addresses this by adding an integrated code editor within the website, helping learners move smoothly from theory to practice. This sub-problem highlights the importance of implementing whatever knowledge you learn.

2.3.4 Engagement and Motivation

Keeping learners engaged and motivated is an important task in online education. Passive learning of content can lead to boredom and reduced learning outcomes. EduVerse aims to encourage engagement by introducing competition-based learning activities that will help learners to test their knowledge and skills in an healthy environment. EduVerse seeks to address the issue of learner motivation by some friendly quizzes and competitions.

2.4 Goals and Objectives

The main goals of EduVerse include:

- Personalized Learning Paths: To custom make learning journeys based on individual strengths

and weaknesses, guiding learners toward their interests and future career prospects.

- Precise Performance Breakdowns: Giving recruiters more power by providing them with detailed insights into an applicant's skills.
- Smooth Learning Experience: To offer an integrated code editor within the platform, ensuring a smooth transition from theory to practice.
- Competition-Based Learning: To support engagement by enabling learners to test their knowledge through competition-based activities.

2.5 Project Scope

EduVerse presents a progressive scope, aiming to better the functionalities of renowned educational platforms into one website. At its core, the platform will analyze individual strengths and weaknesses to create tailored learning paths. It will also incorporate a built-in code editor, encourage competition-based learning, and help the recruitment process.

2.6 Sustainable Development Goal (SDG)

In accordance with the SDG of 'Quality Education,' the EduVerse project aims to make significant changes in enhancing the quality of educational resources. Our platform aims to breaking down barriers to quality education by providing a personalized learning experience to every learner, chance to implement their knowledge, and competition-based learning activities. This approach gives a deeper understanding of subjects and promotes critical thinking and problem-solving abilities. Goal for 'Quality Education' makes EduVerse equip learners with the skills and knowledge they need to succeed in a rapidly changing world. The list of SDGs are given below for reference in Figure 2.1.



Figure 2.1: Quality Education

This figure represents the target Sustainable Development Goal for focused on Quality Education.

2.7 Constraints

Despite our ambitious goals and commitment to achieving sustainable development in the field of online education, the EduVerse project acknowledges the presence of certain limitations that must be addressed first. Firstly, we will address financial limitations that may affect the speed of development and scalability. Budgets, funding, and resource allocation are important to ensure the project's sustainability. Secondly, technological constraints may arise, particularly as the project advances and requires a polished infrastructure. Ensuring the platform's accessibility to users with varying levels of technological resources is a challenge we actively see. Thirdly, regulatory and compliance issues within the education and publishing sectors may pose hurdles. Adhering to legal and ethical standards while fostering innovation is a complex balance that demands careful attention. Lastly, user adoption and engagement are critical for success, and overcoming the inertia of existing educational systems and platforms can be challenging. Despite these constraints, EduVerse remains committed to its mission of enhancing the quality of education and forging a path toward achieving Sustainable Development Goals.

2.8 Business Opportunity

EduVerse is not only an online education platform but a business opportunity as well. EduVerse has the potential to attract learners, institutions, and recruiters, in short the educational industry at large. Addressing the problem of lack of personalization in our educational methods, we have a niche market to get into.

2.9 Stakeholders Description/ User Characteristics

Here we will identify and explain the various stakeholders of EduVerse, and define their respective roles and responsibilities. Afterwards, we will summarize the goals and objectives of our stakeholders and the problems we plan on addressing within our organization.

2.9.1 Stakeholders Summary

The Eduverse platform deals with a diverse range of stakeholders, each playing a crucial role in our platform's mission to transform educational methods. Our stakeholders include:

1. **Learners (Applicants)**: Learners are the main users of our platform, they look for a personalized learning experience and skill development which is made just for them.
2. **Administrators (Admins)**: Admins are responsible for managing the platform, ensuring it runs

smoothly, and organizing settings for learners and educators.

3. **Educators/Instructors**: Stakeholders add the educational content, courses, and resources to EduVerse. They play an important role in shaping the learning experiences offered on the platform
4. **Recruiters and Interviewers**: HR's and interviewers will use EduVerse to dive deep into learners skills and abilities.
5. **Educational Institutions**: Institutions like FAST-NUCES can use EduVerse to run their recruitment processes, improve education quality, and enhance the learning experiences of their students.

2.9.2 Key High-Level Goals and Problems of Stakeholders

The stakeholders within the EduVerse platform have respective goals and problems they aim to address, including:

- **Learners (Applicants)** seek personalized learning paths and practical skills development.
- **Administrators (Admins)** aim to manage and customize the platform efficiently.
- **Educators/Instructors** aspire to deliver high-quality educational content.
- **Recruiters and Interviewers** strive for efficient candidate assessment.
- **Educational Institutions** aim to streamline recruitment and enhance education quality.

Understanding these high-level goals and problems guides the development and optimization of the EduVerse platform, ensuring it aligns with the needs and aspirations of its diverse user base.

2.10 Conclusion

Chapter 2 has provided an overview of the project's vision, identified the problem domain, stated the problem, elaborated on its significance, outlined goals and objectives, defined the project's scope and constraints, and highlighted the business opportunity it presents. This chapter sets the stage for a comprehensive exploration of EduVerse's development and implementation in the subsequent chapters.

Chapter 3 Literature Review / Related Work

This chapter explores the existing research, software, and educational strategies that influenced the creation of EduVerse. A thorough investigation of earlier work in the areas of online learning, individualized learning, and the integration of technology in education served as the basis for EduVerse. We hope to expand upon and improve the ideas and understandings discovered in the following works.

3.1 Definitions, Acronyms, and Abbreviations

List all important definitions, acronyms, and abbreviations used in this chapter.

B Corp: Benefit for All Corporation

K-12: K-12, a term used in education and educational technology in the United States, Canada and some other countries, is a short form for the publicly supported school grades prior to college. These grades are kindergarten (K) and first through 12th grade (1-12).

3.2 Detailed Literature Review

This section will be discussing similar applications in detail by doing a critical analysis on each of them and their relationship to our proposed research work.

3.2.1 Coursera

Coursera [1] is an online learning network which provides a huge selection of classes, specializations, and degree programs in many different subject areas. Coursera collaborates with more than 275 top universities and businesses, to give students around the world high-quality educational content, by providing a variety of learning options, from practical projects and classes to degree programs and certificates for those who are ready for employment.

3.2.1.1 Summary

Daphne Koller and Andrew Ng launched Coursera in 2012 with an ambitious vision to change the education experience and provide educational opportunities to students all over the world. Coursera is an international platform for online education and professional development. They offer learners free and paid courses which gets you a certificate to showcase your skills to the world. They further offer fully accredited courses of bachelors and masters in many fields, by affiliating with renowned universities. Their vast collection of courses encourage students to learn new things which might interest them.

3.2.1.2 Critical analysis of the research item

The development of Coursera has many notable strengths. Coursera aims to provide remarkable learning resources to students worldwide and they have most likely accomplished this. It has crossed borders with its online platform, giving students from various backgrounds the chance to receive education that could change their lives. Its extensive course selection is one of Coursera's best qualities. This collection makes sure that students may select classes that match their interests and professional goals. Additionally, Coursera's collaboration with renowned organizations around the world are proof of the quality and reliability of its course content. These partnerships have improved Coursera's standing and given students confidence in the worth and reliability of the certifications they earn. Although Coursera is a powerful tool in online education, it is not without its challenges. A significant issue is the price of earning certifications for courses. Although the site offers free courses, certification often requires a fee, which for some learners may be out of their price range. In contrast to typical classroom settings, involvement with instructors and students may be restricted due to the courses being mainly just online material. Some students might miss the in-person interaction and fast feedback that face-to-face learning situations offer. Another one of the ongoing issues on Coursera is low course completion rates because many students sign up for courses but do not end up finishing them. The failure of Coursera to get its credentials recognized in the job market is a problem as well. Even though its certifications are extremely beneficial for developing abilities, some organizations might not fully accept them as being comparable to conventional degrees just yet. Although this attitude is gradually changing, Coursera students may still be impacted in terms of job possibilities and professional advancement.

3.2.1.3 Relationship to the proposed research work

The relationship between Coursera and EduVerse lies in the shared domain of online education and e-learning platforms. While they are distinct entities, they both operate within the field of digital education, targeting similar objectives of providing accessible and effective learning experiences to a global audience. EduVerse can be seen as a potential competitor or an alternative to Coursera. As EduVerse aims to merge the strengths of various e-learning platforms, including Coursera, it may offer a similar catalog of courses and features. Learners evaluating their options for online education may consider both platforms. But on the other hand, EduVerse also aims to differentiate itself by focusing on personalized learning paths, performance insights, integrated coding environments, and competition-driven learning. These unique features set it apart from Coursera and other e-learning platforms.

3.2.2 Educative

Educative [2] is a dynamic online learning platform that has gained prominence recently in the field of software development and technology education. Their mission is to change the way individuals learn and master technical skills. Educative offers a diverse and comprehensive collection of courses and tutorials to fulfill their aim.

3.2.2.1 Summary

Educative was founded in 2015 by Fahim ul Haq with the intention of helping developers become more efficient and productive. His goal was to establish an interactive learning platform that didn't require development environments to be set up, installed, or configured. Interactive and hands-on learning methods are used in educational settings. Unlike traditional text-based or video lectures, Educative's platform offers students a unique coding environment that enables real-time practice and application of coding principles, hence enhancing learning.

3.2.2.2 Critical Analysis of the Research Item

Educative is a powerful online education platform because of a number of reasons. Its unique hands-on learning approach is a noteworthy attribute. Without having to set up or download any settings, students may apply what they've learned in real-time due to their interactive coding environment. This practical component strengthens understanding and skill development, which makes it especially useful for technical disciplines where practical experience is important. The platform's wide course collection is yet another important feature. Educative offers a vast collection of technical topics carefully chosen by professionals in the field. On the other hand, Educative's focus on technical subjects is one of its major drawbacks. This may limit its attractiveness to students who are interested in non-technical areas. Furthermore, while many find Educative's hands-on approach to be a strength, not all students may benefit from it. Certain people could have a preference for standard, text-based learning approaches or find the coding environment difficult to navigate. This could result in a learning curve that may discourage certain users. Moreover, since Educative emphasizes practical application, it may lack the theoretical depth required for some advanced topics. As such, learners seeking a more comprehensive theoretical understanding may need to supplement their education with additional resources.

3.2.2.3 Relationship to the proposed research work

Educative and EduVerse are two organizations within the field of online education and e-learning platforms, each with a distinct focus and goals. Although they work in the same field, their objectives and

target audiences are separate. Both are resources that complement one another. As a comprehensive learning platform, EduVerse seeks to combine the advantages of other e-learning systems (including Educative) with its own unique features. Students using EduVerse will have access to courses and a practical approach similar to Educative. Main areas of emphasis for EduVerse include personalized learning routes, performance insights, integrated coding environments, and competition-driven learning. Educative, in contrast, focuses on providing technical education with a focus on practical coding experience. The two platforms aim to achieve different things, even though there might be some material similarity.

3.2.3 Nearpeer

Nearpeer [3] is an innovative educational platform that has changed the way students access academic support and connect with their peers. Nearpeer, which was established with the goal of promoting community-driven academic support and collaborative learning, has become a significant platform in the field of student engagement.

3.2.3.1 Summary

Ammar Ali Ayub, Shahrukh Swati, and Umair Latif created Nearpeer in 2016 with the goal of utilising technology to lower the cost of high-quality education and increase access throughout Pakistan and the Middle East. They believe that everyone should have equal access to opportunities for learning and growth, regardless of their financial status or place of residence.

3.2.3.2 Critical Analysis of the research item

Nearpeer has several significant advantages, among these is peer-to-peer learning, which is done in a creative fashion. Nearpeer provides students with access to valuable resources and support by matching them with classmates who have faced identical academic problems. Through the development of friendships and a sense of community, this approach dramatically improves the quality of learning for all students. Nearpeer offers expert-led workshops and cooperative study groups which is an additional noteworthy feature. These elements promote interactive learning and give students access to subject matter experts. Additionally, the platform's active online community allows students to interact, exchange ideas, and gain access to educational resources. Nearpeer does have certain limits along with its advantages. The platform might be excellent in some areas, but it doesn't offer thorough assistance for all academic subjects. This could be a disadvantage for students looking for help in a specialised or uncommon disciplines. Furthermore, peers' willingness and ability to help one other out is a requirement for the peer-to-peer model of Nearpeer to function well.

3.2.3.3 Relationship to the proposed research work

Nearpeer and EduVerse are two separate businesses each with a specific purpose. Nearpeer puts a strong focus on peer-to-peer interaction to build a sense of shared knowledge and support within a student community. Its main goals are to promote collaborative learning and community-driven assistance among students. In contrast, EduVerse is an online learning platform designed for the development of technical skills. It provides a vast range of tutorials and courses that are primarily targeted at technology enthusiasts and software engineers. Students can opt to use both platforms in order to take advantage of Nearpeer's peer assistance and community involvement as well as EduVerse's skill-building courses to obtain technical knowledge.

3.2.4 HackerRank

HackerRank [4] is an established company that offers ambitious programmers and developers a cutting-edge and dynamic platform. Their goal is to expand coding and making technical education accessible to all. HackerRank has become the go-to resource for enhancing coding abilities, getting ready for technical interviews, and participating in coding competitions. The basic principle of HackerRank is that while skill is natural, opportunities are not. The goal of the platform is to close this gap by providing equal opportunities for all students, irrespective of their geography or background.

3.2.4.1 Summary

HackerRank is a well-known website for coding and technical skill improvement and it has completely changed how people think about software engineering and programming. HackerRank enables students of all skill levels to not only improve their coding skills but also get ready for technical interviews by providing them with a wide range of code challenges, in-depth tutorials, and coding competitions. Its dedication to giving all students equal oppurtunity is a proof of its goal to equalise coding instruction and close the achievement gap.

3.2.4.2 Critical analysis of the research item

HackerRank has a number of distinguishing advantages that have helped it to achieve recognition in the field of technical skill development. Firstly, its vast collection of coding challenges and comprehensive tutorials caters to learners of all levels, from beginners to seasoned developers. This extensive library ensures that users can find content that matches their skill level and allows for continuous skill improvement. Secondly, HackerRank's coding competitions and technical interview preparation resources are highly effective in helping individuals prepare for real-world job opportunities. Its focus on practi-

cal problem-solving and algorithmic thinking aligns with the demands of the tech industry. Despite its strengths, HackerRank does have some limitations. One notable drawback is that its content, particularly in the domain of coding challenges, can be quite challenging for absolute beginners. This may make it less accessible for those who are just starting their coding journey. Additionally, while HackerRank is an excellent resource for coding skills, it may not offer as much depth in terms of broader software development concepts or technologies. Learners seeking comprehensive understanding in areas beyond coding algorithms may need to complement their HackerRank experience with other resources.

3.2.4.3 Relationship to the proposed research work

HackerRank and EduVerse are two different platforms that serve different goals, have different functions, and target different audiences. First and foremost, HackerRank is a tool for refining coding and programming abilities. This is done by offering an array of coding tasks, contests, and evaluations to users. EduVerse, on the other hand, approaches education from a more complete standpoint. Even while it has a built-in code editor, its main goal goes beyond solving coding problems. HackerRank is the best at practicing coding and developing technical skills, whereas EduVerse aims to offer a more inclusive and wide-ranging educational experience. HackerRank and Eduverse's integrated coding environment and competition-based learning are comparable even though the goals and target audiences of these platforms differ.

3.2.5 Codecademy

Codecademy [5] is an innovative and widely recognized online learning platform offering an extensive array of interactive coding courses designed to empower learners of all levels. Codecademy has changed the way individuals learn and apply programming languages and web development skills With a user-friendly interface, hands-on coding experiences, and a commitment to making coding accessible to everyone.

3.2.5.1 Summary

Codecademy was founded in 2011 by Zach Sims and Ryan Bubinski. It stands as a prominent online learning platform dedicated to teaching coding and programming skills. It has transformed the world of coding education by offering a diverse catalog of interactive courses accessible to learners of all levels. The platform's hallmark is its hands-on approach, where users write and execute code directly within the platform. Codecademy's mission is clear, which is to make coding education accessible, engaging, and effective for everyone.

3.2.5.2 Critical Analysis of the research item

Codecademy is a popular online learning platform known for its coding and programming courses. Like any educational platform, it has its own set of strengths and weaknesses. It has a number of qualities including providing an interactive learning experience, allowing users to practice coding directly within its platform. This hands-on approach is a key strength, reinforcing understanding and application of programming languages effectively. The platform's user-friendly interface is especially helpful for beginners, making it very easy for new comers. While Codecademy is a valuable resource, it does have its weaknesses as well. It may not help as effectively to advanced programmers due to its limited depth in advanced topics. The platform tends to prioritize coding practice over teaching the theory of computer science or software engineering, which may be a drawback for those seeking a deeper understanding. Some advanced or specialized courses require a paid subscription which might be a barrier for certain learners.

3.2.5.3 Relationship to the proposed research work

Codecademy and EduVerse are both online learning platforms, but they serve different purposes and have distinct offerings. Codecademy primarily focuses on teaching coding and programming skills through interactive coding exercises and projects. It offers a wide range of courses on programming languages and technologies. In contrast, EduVerse aims to be a comprehensive learning platform offering a personalized learning experience, detailed performance insights, and more. While Codecademy specializes in coding and programming courses, EduVerse's scope is broader. EduVerse intends to provide a one-stop solution for all educational needs. While there may not be many similarities between the two platforms, hands-on coding experience with a coding platform available in both platforms is something that related them together.

3.2.6 Khan Academy

Khan Academy [6] is an innovative platform emerged from a simple idea to provide free, high-quality education to anyone, anywhere. It was established in 2008 by Salman Khan. Since then, Khan Academy has evolved into a strong competitor in education industry. They offer educational resources spanning over many subjects, from mathematics and science to the arts and humanities.

3.2.6.1 Summary

Khan Academy is an influential online educational platform designed with the fundamental purpose of providing free, world-class education to anyone with an internet connection. This platform changed

traditional learning methods by offering an extensive library of instructional videos, practice exercises, and educational resources across various subjects. Khan Academy's mission is to democratize education, making it accessible to learners of all ages and backgrounds, regardless of geographical location or financial constraints. Khan Academy empowers individuals to explore, comprehend, and excel in subjects ranging from mathematics and science to the humanities.

3.2.6.2 Critical analysis of the research item

Khan Academy's primary strength lies in its commitment to providing free, high-quality educational content to a global audience. Its extensive library of instructional videos and practice exercises covers a wide range of subjects, making it an invaluable resource for learners of all ages. The platform's self-paced learning approach and clear, concise explanations empower individuals to master complex concepts at their own speed. On the contrary, Khan academy does have some limitations. One of Khan Academy's notable weaknesses is its limited depth in certain subjects and topics. While it excels in providing foundational knowledge and practice exercises, it may not offer the same level of depth and complexity as some advanced educational platforms or traditional higher education institutions.

3.2.6.3 Relationship to the proposed research work

Khan Academy and EduVerse, despite their distinct purposes and target audiences, share some similarities. Khan Academy primarily focuses on K-12 education and foundational subjects, both platforms cover a diverse array of subjects. EduVerse offers a broader range of topics, including academic subjects, professional development, coding, and more. Both platforms promote self-paced learning, allowing individuals to progress through courses and content at their own speed. This flexibility accommodates a wide range of learning styles and preferences. While Khan Academy offers resources for teachers and educators to integrate its content into their classrooms, EduVerse's personalized learning paths and performance insights could also benefit educators in tailoring their teaching approaches to individual student needs. Despite these similarities, it's important to note that the primary distinctions between the two platforms lie in their specific goals, target audiences, and the depth of content offered. Khan Academy is renowned for its focus on foundational education, particularly for K-12 students, while EduVerse seeks to offer a more comprehensive educational experience that extends beyond K-12 and caters to a diverse range of learners and educational needs.

3.2.7 Linkedin Learning

LinkedIn Learning [7], a prominent and dynamic online education platform, represents a transformative approach to professional development and lifelong learning. Rooted in the vision of enhancing career

prospects and fostering continuous growth, LinkedIn Learning offers a wealth of courses designed to empower individuals across various industries.

3.2.7.1 Summary

LinkedIn Learning stands as a premier online educational platform with a primary mission: to equip individuals with the skills and knowledge needed to excel in their careers and adapt to an ever-evolving professional landscape. Offering a vast and diverse catalog of courses, LinkedIn Learning empowers users to explore topics ranging from business and technology to creative arts and soft skills. Its purpose is to provide accessible, on-demand learning experiences that align with career goals and personal development aspirations. What sets LinkedIn Learning apart is its seamless integration with the LinkedIn network, allowing users to showcase their completed courses and certifications on their LinkedIn profiles, thus enhancing their professional visibility and networking opportunities.

3.2.7.2 Critical Analysis of the research item

LinkedIn Learning's standout strength lies in its integration with the LinkedIn professional network. This seamless connection allows learners to display their completed courses and certifications on their LinkedIn profiles, effectively showcasing their skills to potential employers and collaborators. Additionally, its vast course library covers an extensive range of topics, ensuring relevance for a wide audience. The quality of instruction is typically high, with courses often led by industry experts. This platform's accessibility, with the ability to learn at one's own pace and from anywhere, makes it an invaluable tool for professionals seeking to upskill or advance their careers.

LinkedIn Learning does have some limitations, notably related to its pricing model. While it offers a free trial period, continued access to its extensive course library requires a subscription, which may be costly for individual users. Additionally, the platform primarily focuses on skills-based courses, potentially lacking the depth and breadth of content found in traditional degree programs or specialized institutions. The platform's effectiveness also depends on the learner's motivation and self-discipline, as the absence of structured classes and deadlines may lead to incomplete courses. Finally, while the LinkedIn integration is a strength, it may also raise privacy concerns for those who wish to keep their professional and educational pursuits separate

3.2.7.3 Relationship to the proposed research work

LinkedIn Learning and EduVerse share several notable similarities. Both platforms are dedicated to online learning, offering users the convenience of acquiring new skills and knowledge from anywhere in the world. They boast extensive course catalogs spanning various subjects, accommodating learn-

ers with diverse interests and educational objectives. Primarily focused on professional development, both LinkedIn Learning and EduVerse empower individuals to enhance their career prospects and personal growth. Additionally, they promote self-paced learning, enabling users to tailor their educational journeys to their schedules and preferences. These shared characteristics highlight the potential synergy between the platforms, with LinkedIn Learning's courses seamlessly complementing EduVerse's broader educational ecosystem, providing learners with a well-rounded and personalized educational experience.

3.3 Related Work Summary Table

The related work summary table is shown in the table 3.1.

Table 3.1: Related Work Review Summary Table

This summary Table summaries the points discussed throughout this chapter

Application	**Features**	**Relevance to Application**	**Limitations**
Coursera [1]	Diverse collection of online courses	Educational content	No certification without paying
Educative [2]	Hands-on learning approach	interactive coding tutorials	Limited content outside technical subjects
Nearpeer [3]	Peer-to-Peer Learning.	Collaborative learning communities	Limited course offering
HackerRank [4]	Competition Based Learning. Coding Practices. Technical job interview preparations.	Integrated coding environment and coding challenges and competitions	Coding challenges can be tough for beginners. No hands on project work experience
Codecademy [5]	Interactive Coding Exercises. Hands-on Coding Projects	Integrated coding environment and interactive coding exercises	Limited depth in advanced topics
Khan Academy [6]	Education Video Lessons	Educational video and written content	Limited depth in certain subjects and topics
Linkedin Learning [7]	Integration with LinkedIn Profile. High-Quality Video Courses	Educational Content	Costly for individual users

3.4 Conclusion

In conclusion, this chapter has explored the intricate relationships between EduVerse and several prominent online learning platforms, namely Coursera, Educative, Nearpeer, HackerRank, Codecademy, Khan Academy, and LinkedIn Learning. Each platform offers unique features and resources that can inspire and enrich EduVerse's educational ecosystem. EduVerse aims to leverage the strengths of these platforms to create a comprehensive and self-contained learning environment.

Chapter 4 Software Requirement Specifications

This chapter highlights important features of the projects. It also includes functional and non-functional requirement of the project, defines use cases and risk analysis involved in the project.

4.1 List of Features

1. **Personalized Learning Paths:** Users can create personalized learning paths based on their skills, interests, and career goals. The system suggests courses, tutorials, and learning resources tailored to each learner's needs.
2. **Skill Assessment and Gap Analysis:** EduVerse conducts skill assessments to identify strengths and weaknesses in a learner's knowledge. It then provides a detailed gap analysis, recommending specific topics or courses to address these gaps.
3. **Code Editor and Practice Environments:** Integrated code editors for various programming languages and technologies allow learners to practice coding, algorithms, and problem-solving in a real coding environment without switching to external tools.
4. **Competitive Learning:** Implement a gamified approach where learners can participate in coding challenges, quizzes, and competitions to test their skills against others, earn badges, and climb leaderboards.

4.2 Functional Requirements

This section will be discussing functional requirements for the different entities of EduVerse.

4.2.1 Functional Requirements for Instructors:

Instructors within EduVerse shall be able to:

1. **Create and Manage Courses:** The system shall allow instructors to create new courses, upload course materials, and manage course content.
2. **Schedule Live Sessions:** The system shall enable instructors to schedule live online sessions, webinars, or workshops, providing details such as date, time, and access links.
3. **Monitor Learner Progress:** The system shall allow instructors to track learner progress within their courses, view assignment submissions, and provide feedback to learners.
4. **Conduct Assessments:** The system shall provide functionality for instructors to create quizzes, exams, and assignments, set grading criteria, and review and grade them accordingly.

5. **Access Analytics:** Instructors shall be able to access analytics and reporting tools within the system to analyze learner engagement, course performance, and other relevant data.
6. **Publish and Update Course Content:** Instructors shall have the capability to publish, update, or archive course content to ensure learners have access to the most current materials.

4.2.2 Functional Requirements for Learners:

Learners using EduVerse shall be able to:

1. **Create User Account:** Learners shall be able to create a user account with a unique profile, including personal information and learning preferences.
2. **Browse Courses:** The system shall enable learners to browse and search for courses based on their interests, goals, or specific topics.
3. **Enroll in Courses:** Learners shall have the ability to apply for enrollment in courses of their choice, access course materials, and receive notifications regarding course updates.
4. **Participate in Live Sessions:** Learners shall be able to join live online sessions, webinars, or workshops scheduled by instructors.
5. **Complete Assignments and Assessments:** Learners shall have the capability to submit assignments, take quizzes, and participate in course assessments.
6. **Track Progress:** The system shall allow learners to monitor their course progress, view grades and feedback, and access completion certificates.
7. **Access Learning Resources:** Learners shall have access to a repository of learning resources, including videos, articles, and supplementary materials.
8. **Participate in Competitions:** Learners shall be able to participate in competitions relevant to their course material.

4.2.3 Functional Requirements of the System:

The EduVerse system shall be capable of:

1. **User Authentication:** The system shall provide secure user authentication, allowing learners and instructors to create accounts and log in securely.
2. **Content Management:** The system shall facilitate the management of course content, including storage, retrieval, and updates.
3. **Personalization:** The system shall offer personalized learning experiences, recommending courses,

materials, and resources based on user preferences and performance.

4. **Assessment and Grading:** The system shall support the creation and management of assessments, automated grading, and feedback distribution.
5. **Communication:** The system shall enable communication between users, including messaging, notifications, and email alerts.

4.3 Quality Attributes

Ensuring the success and effectiveness of the EduVerse platform necessitates a careful consideration of various quality attributes that define its performance, reliability, and overall user experience. These attributes, crucial for the platform's functionality and long-term sustainability, encompass factors such as performance to guarantee seamless user interactions, reliability for consistent availability, usability for an intuitive and accessible interface, and security to protect user data and privacy. Additionally, considerations extend to maintainability, scalability, and adaptability to accommodate future enhancements and a growing user base. This section delineates and elaborates on the key quality attributes that underpin the design and development of EduVerse, aiming to deliver an exceptional and enduring educational experience.

4.4 Non-Functional Requirements

This section will be discussing non-functional requirements for the different entities of EduVerse

4.4.1 Performance

The system should meet the following performance requirements:

- Users must be able to access the site within 5-10 seconds.
- The system should be efficient enough to handle a realistic number of users simultaneously.

4.4.2 Reliability

The application must ensure maximum possible up-time, with the goal of achieving at least 99% up-time.

4.4.3 Usability

The system should provide a user-friendly interface that allows users to use the application within minutes, and it should be self-explanatory in its operation.

4.4.4 Re-usability

Code Re-usability: The code should be organized into modules based on functionality to promote code re-usability. Modular code design allows for easier reuse of code components across different parts of the application.

4.5 Assumptions

The following assumptions have been made for this system:

- End users have a web browser installed on their desktop that is JavaScript compatible.
- As the web application will be deployed for the users, an internet connection is a prerequisite.
- Users can understand the basics of the website, including navigation, buttons, and other user interface elements.

4.6 Hardware and Software Requirements

This section will be discussing the hardware and softawre requirements that are necessary to develop the web application EduVerse.

4.6.1 Hardware Requirements

The following hardware requirements are necessary for the project:

- Desktop PC/Laptop
- Internet connection is required.

4.6.2 Software Requirements

The following software requirements are necessary for the project:

- Vs Code
- React JS
- Firebase
- Ace.js

Table 4.1: Use Case 1: User Registration

A new user creates an account on EduVerse.

Name	User Registration		
Actors	New User		
Summary	A new user creates an account on EduVerse.		
Pre-Conditions	None		
Post-Conditions	User account is created.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	User accesses the registration page.	2	The system displays the registration form.
2	User provides personal information.	3	System records the provided information.
3	User selects a username and password.	4	System verifies the uniqueness of the username.
4	User completes the registration process.	5	System creates the user account.
Alternative Flow			
3	User selects a username that is not unique.	4-A	System responds with an error message: Username already exists.

4.7 Use Cases

Use cases in the context of a system architecture play a crucial role in defining, designing, and validating the functionality and structure of the final system. They represent significant scenarios or interactions that drive the development and operation of the system.

The table 4.1 onwards provide the use cases for the project EduVerse.

Table 4.2: Use Case 2: User Login

A registered user logs in to their EduVerse account.

Name	User Login		
Actors	Registered User		
Summary	A registered user logs in to their EduVerse account.		
Pre-Conditions	User has a registered account.		
Post-Conditions	User is logged in.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	User accesses the login page.	2	The system displays the login form.
2	User enters their username and password.	3	System validates the credentials.
3	User submits the login form.	4	System grants access upon successful validation.
Alternative Flow			
2	User enters incorrect username or password.	3-A	System responds with an error message: Invalid credentials.

Table 4.3: Use Case 3: Course Enrollment Request

A learner requests to enroll in a course on EduVerse.

Name	Course Enrollment Request		
Actors	Learner		
Summary	A learner requests to enroll in a course on EduVerse.		
Pre-Conditions	User is logged in.		
Post-Conditions	Enrollment request is submitted.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	Learner searches for a course of interest.	2	The system displays the course search functionality.
2	Learner selects the desired course.	3	System shows course details.
3	Learner clicks the ”Enroll” button.	4	System records the enrollment request.

Table 4.4: Use Case 4: Creating a New Course

An instructor creates a new course on EduVerse.

Name	Creating a New Course		
Actors	Instructor		
Summary	An instructor creates a new course on EduVerse.		
Pre-Conditions	Instructor is logged in.		
Post-Conditions	New course is created and published.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	Instructor accesses the course creation page.	2	The system displays the course creation form.
2	Instructor provides course details and content.	3	System records the course information.
3	Instructor publishes the course for learners to access.	4	System makes the course available to learners.

Table 4.5: Use Case 5: Scheduling a Live Session

Instructors schedule live sessions or webinars.

Name	Scheduling a Live Session		
Actors	Instructor		
Summary	Instructors schedule live sessions or webinars.		
Pre-Conditions	Instructor is logged in, Instructor has selected a relevant course		
Post-Conditions	Live session is scheduled.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	Instructor accesses the scheduling tool.	2	The system displays the scheduling options.
2	Instructor sets the date, time, and topic of the session.	3	System records the session details.
3	Learners view the link, date, time, and topic.	4	System provides the learners with the date, time, topic, and join link.

Table 4.6: Use Case 6: Uploading Course Materials

Instructors upload course materials such as lectures, slides, and assignments.

Name	Uploading Course Materials		
Actors	Instructor		
Summary	Instructors upload course materials such as lectures, slides, and assignments.		
Pre-Conditions	Instructor is logged in, Instructor has selected a relevant course		
Post-Conditions	Course materials are uploaded and accessible.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	Instructor accesses the course content management.	2	The system displays the content management interface.
2	Instructor uploads files in the course.	3	System stores and organizes the uploaded materials.

Table 4.7: Use Case 7: Managing Account Settings

Users can update their account settings, including email preferences and notifications.

Name	Managing Account Settings		
Actors	User		
Summary	Users can update their account settings, including email preferences and notifications.		
Pre-Conditions	User is logged in.		
Post-Conditions	Account settings are updated.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	User accesses the account settings page.	2	The system displays the account settings options.
2	User modifies preferences and saves changes.	3	System updates the user's account settings.

Table 4.8: Use Case 8: Issuing Certificates

Instructors can issue completion certificates to learners who successfully finish a course.

Name	Issuing Certificates		
Actors	Instructor		
Summary	Instructors can issue completion certificates to learners who successfully finish a course.		
Pre-Conditions	Instructor is logged in.		
Post-Conditions	Certificates are issued to eligible learners.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	Instructor accesses the certificate issuance tool.	2	The system displays the certificate issuance options.
2	Instructor/admin selects eligible learners and issues certificates.	3	System generates and issues certificates to selected learners.

Table 4.9: Use Case 9: Quiz

Learners and instructors take quizzes.

Name	Quiz		
Actors	Learner/Instructor		
Summary	Learners take quizzes regarding the material they studied		
Pre-Conditions	User is logged in.		
Post-Conditions	Quiz competition results are recorded.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	User accesses the quizzes list.	2	The system displays the quiz options.
2	User participates in the quiz.	3	System records the quiz results.

4.8 Graphical User Interface

This section should give the GUI dumps of each screen, with reference to the users.

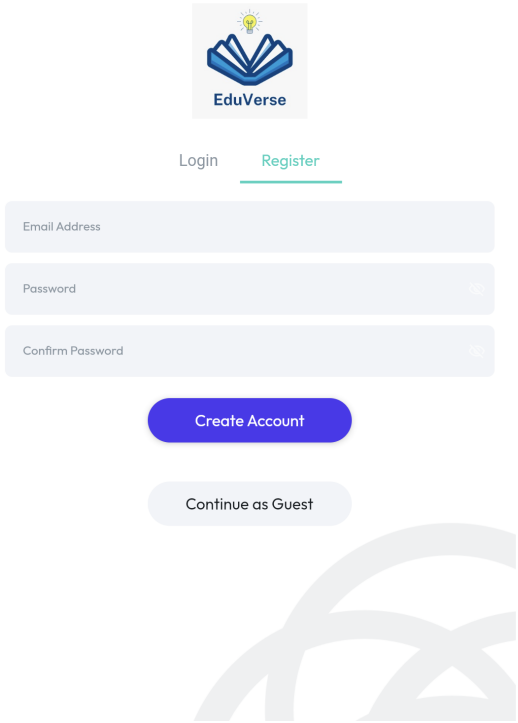


Figure 4.1: Sign-Up Screen (Figure 1)
This screen is an example of EduVerse's Sign-Up Screen, where users can create a new account to access the platform.

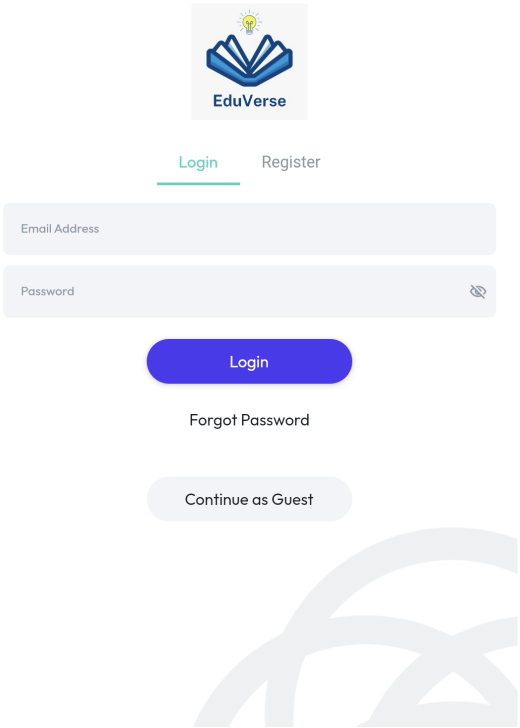


Figure 4.2: Login Screen (Figure 2)
This screen is an example of EduVerse's Login Screen, allowing registered users to log in and access their accounts.

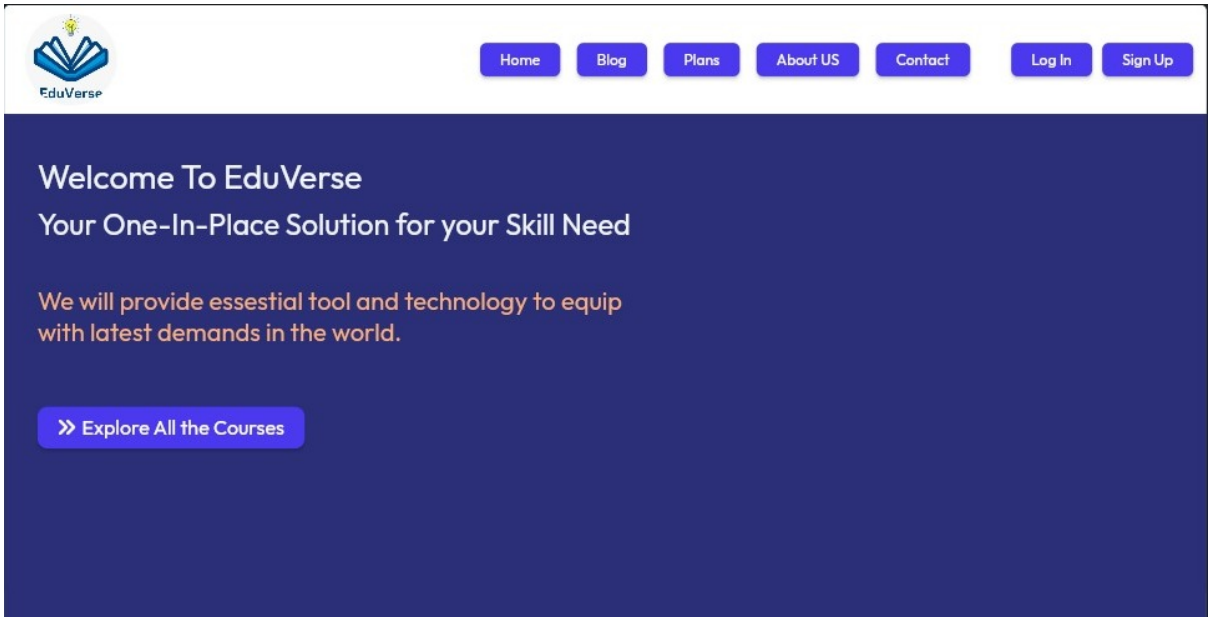


Figure 4.3: Main Page - Part 1 (Figure 3)
This screen represents EduVerse's Main Page.

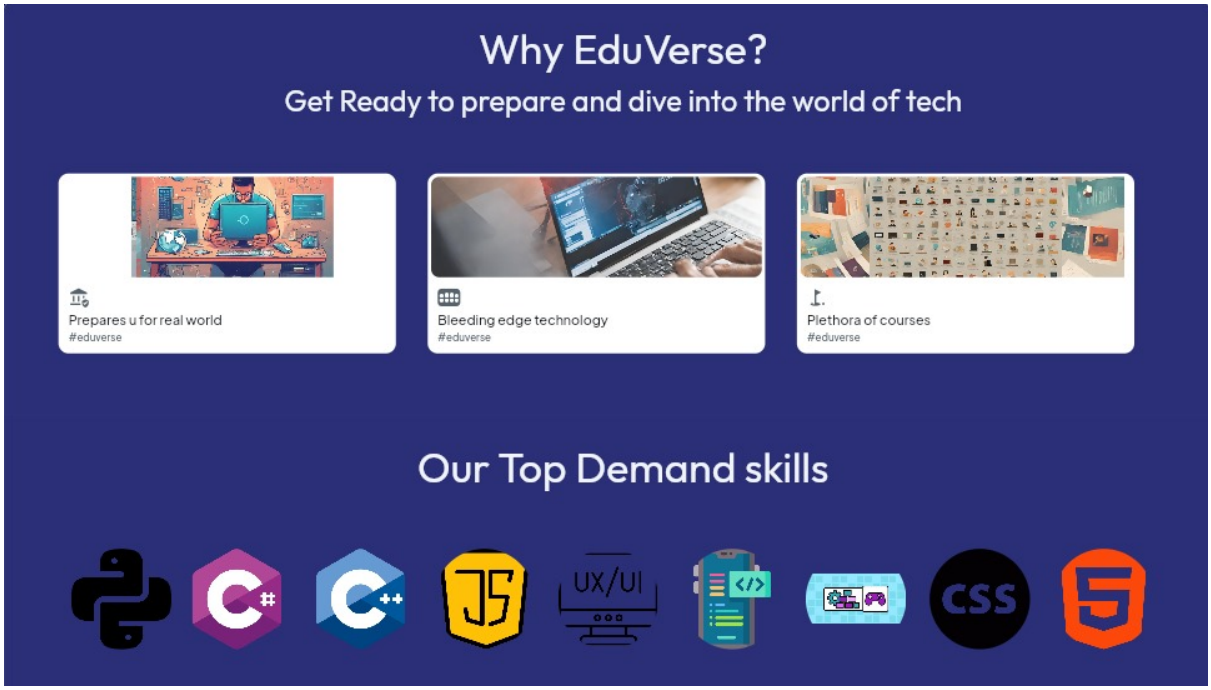


Figure 4.4: Main Page - Part 2 (Figure 4)
The second part of EduVerse's Main Page represents the scrolled view of the main page.

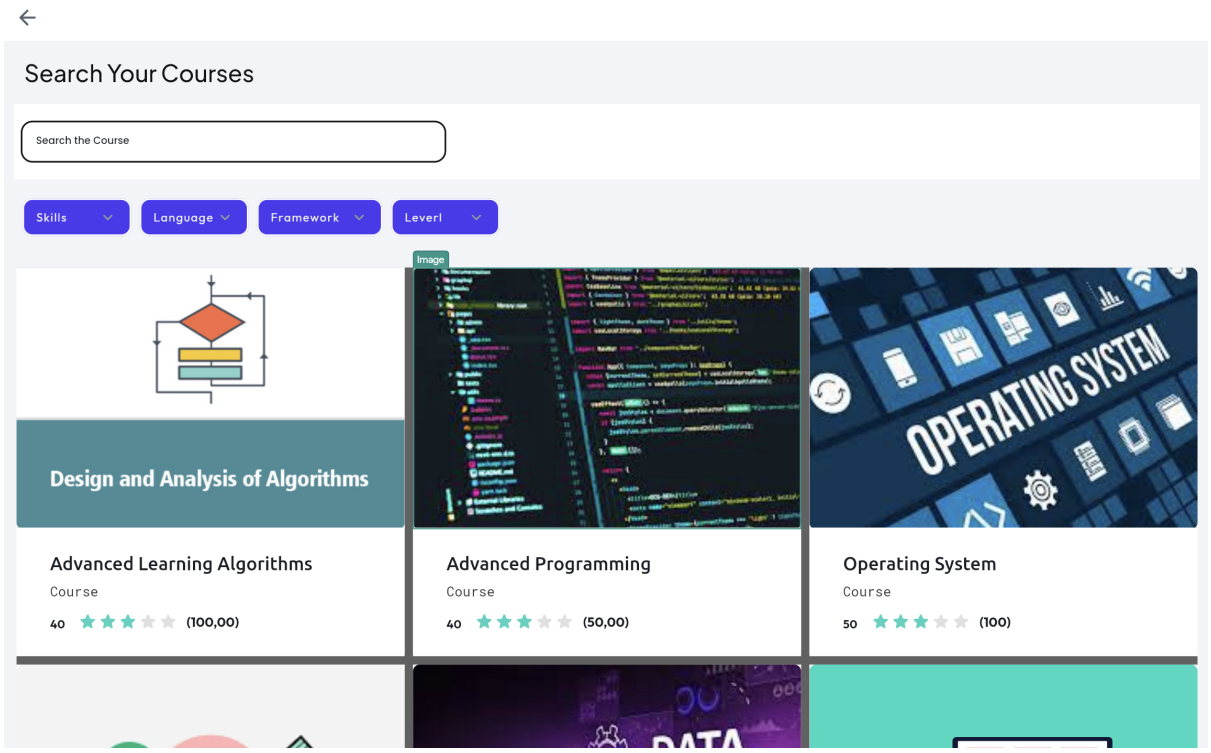


Figure 4.5: Courses Screen (Figure 5)
Users can browse and discover available courses on EduVerse through this Courses Screen.

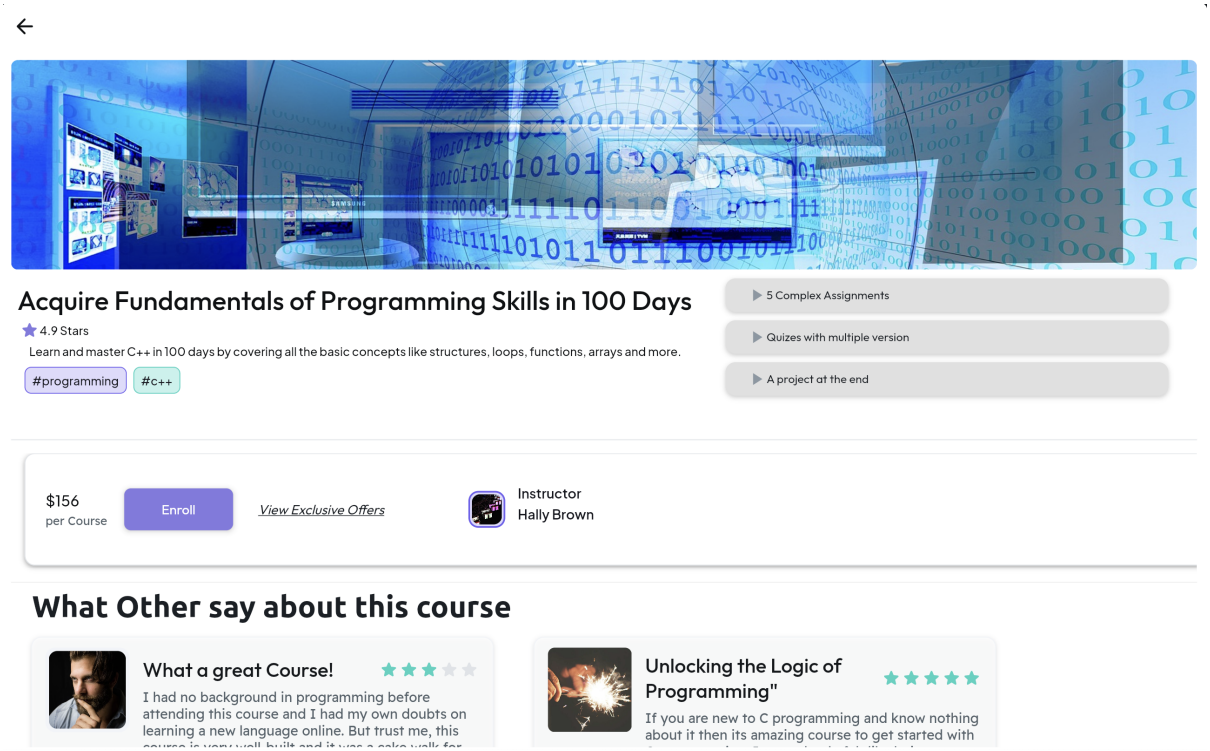


Figure 4.6: Course Description Screen (Figure 6)

This screen displays detailed information about a specific course, helping users make informed decisions.

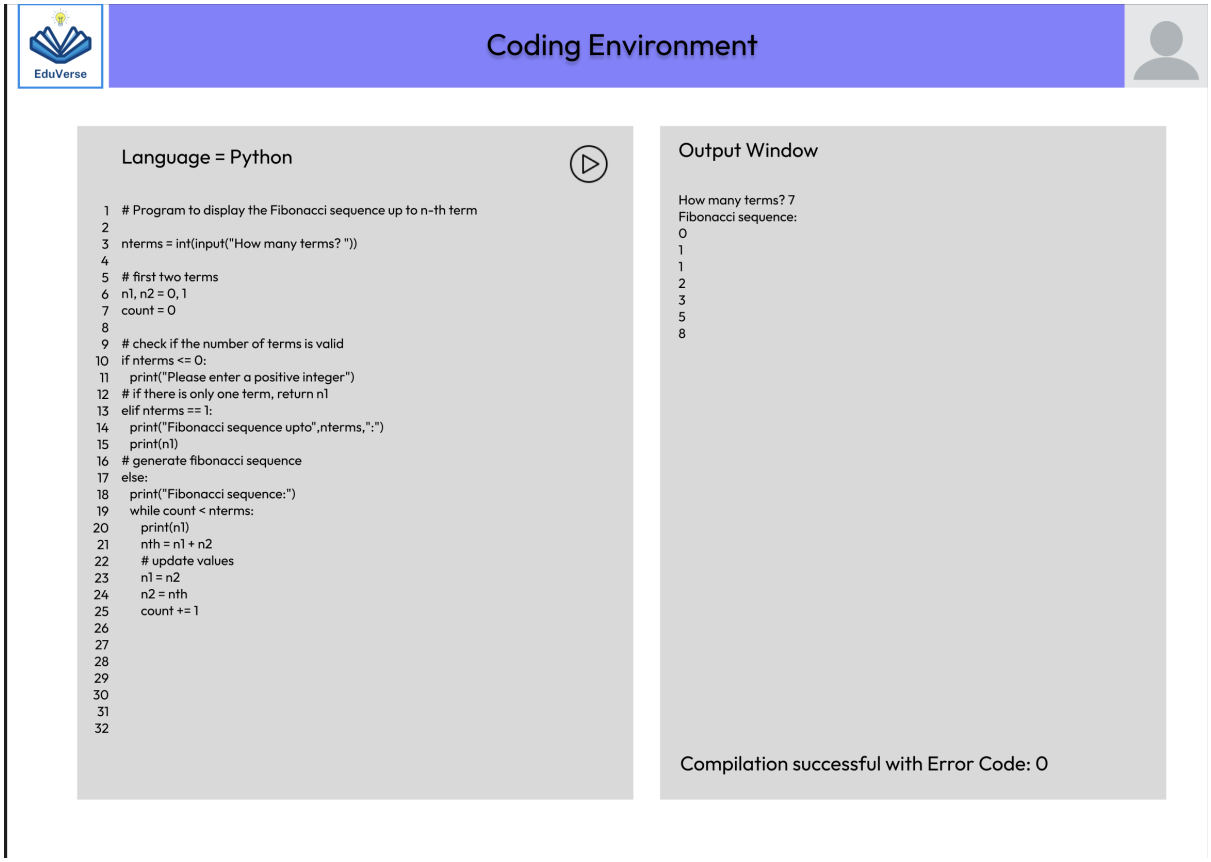


Figure 4.7: Coding Environment Screen (Figure 7)

EduVerse's Coding Environment Screen is where users can practice coding and complete coding exercises.

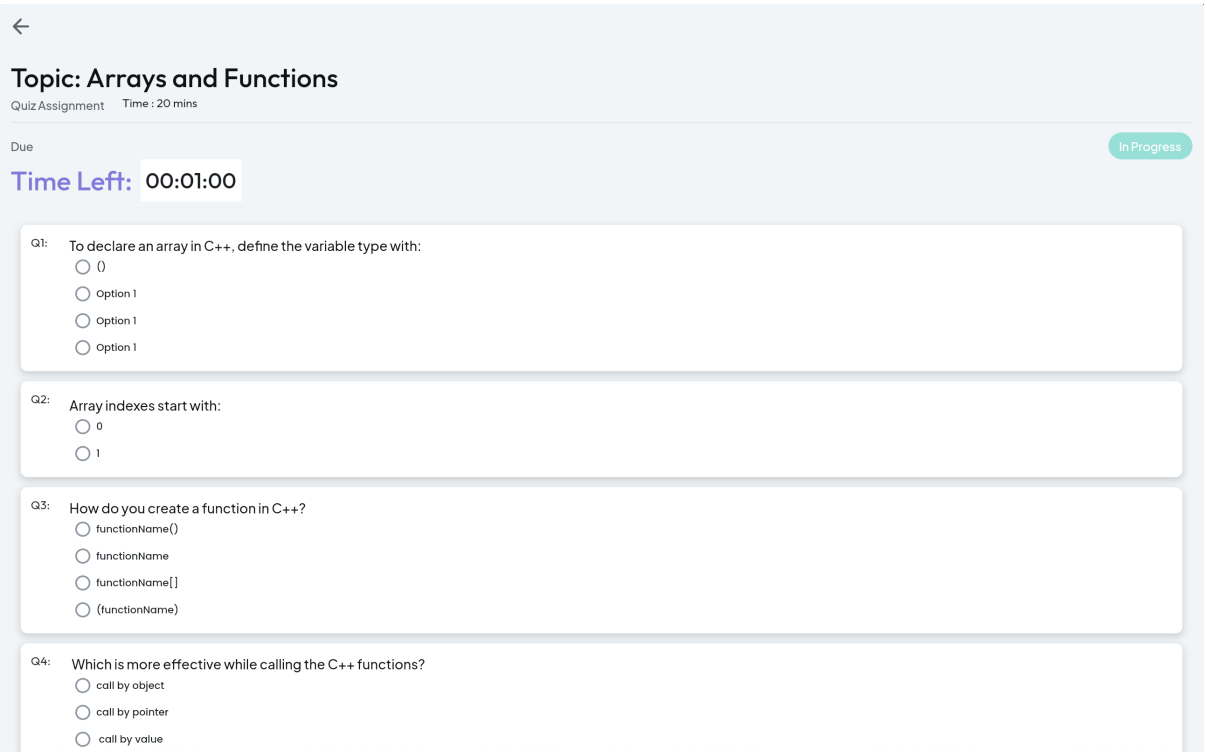


Figure 4.8: Quiz Screen (Figure 8)

Users can take quizzes and assess their knowledge on various subjects through EduVerse's Quiz Screen.



Figure 4.9: Certificate Screen (Figure 9)

This screen showcases the certificates users can earn upon completing courses.

4.9 Database Design

This section will be showcasing the Entity Relationship diagram and the data dictionary associated with it.

4.9.1 ER Diagram

The Entity Relationship Diagram of EduVerse is as follows

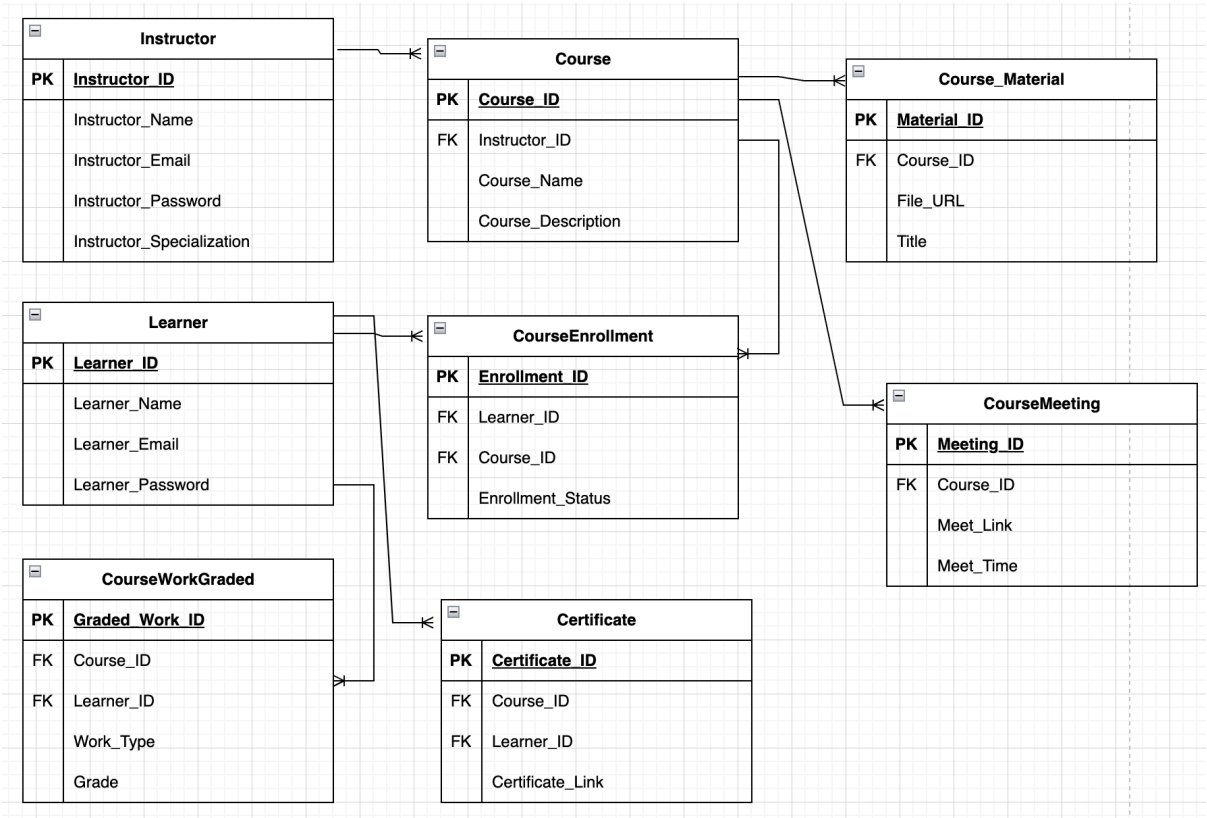


Figure 4.10: Entity Relationship Diagram for EduVerse

Entity-Relationship (ER) Diagram illustrating the relationships among tables including Instructor, Learner, Course, CourseEnrollment, CourseMaterial, CourseWorkGraded, Certificate, and CourseMeeting, detailing entities, attributes, and their connections.

4.9.2 Data Dictionary

The Data Dictionary for each of the entity meantioned in the Entity Relationship diagram is given in this chapter to better visualize the entities and their relations with other entities.

Table 4.10: Instructor Table

This is the sample table for the Instructor Table, detailing entities, attributes, relations, and descriptions.

Entity	Attribute	Data Type	Nullable	Relation to	Relation Type	Description
Instructor	InstructorID	Primary Key	No			Unique identifier for an instructor.
	FirstName	Text	Yes			First name of the instructor.
	LastName	Text	Yes			Last name of the instructor.
	Email	Text	Yes			Email address of the instructor.
	HireDate	Date	Yes			Date of instructor's hiring.
	Specialization	Text	Yes			Area of specialization of the instructor.

Table 4.11: Learner Table

This is the sample table for the Learner Table which includes entities, attributes, relations, and descriptions.

Entity	Attribute	Data Type	Nullable	Relation to	Relation Type	Description
Learner	LearnerID	Primary Key	No			Unique identifier for a learner.
	FirstName	Text	Yes			First name of the learner.
	LastName	Text	Yes			Last name of the learner.
	Email	Text	Yes			Email address of the learner.
	Enrollment Date	Date	Yes			Date when the learner enrolled.
	Program	Text	Yes			Program the learner is enrolled in.

Table 4.12: Course Table

This is the sample table for the Course Table which includes entities, attributes, relations, and descriptions.

Entity	Attribute	Data Type	Nullable	Relation to	Relation Type	Description
Course	CourseID	Primary Key	No			Unique identifier for a course.
	CourseName	Text	Yes			Name of the course.
	Description	Text	Yes			Description of the course.
	StartDate	Date	Yes			Start date of the course.
	EndDate	Date	Yes			End date of the course.
	InstructorID	Foreign Key	Yes	Instructor	One-to-Many	Instructor teaching the course.

Table 4.13: CourseEnrollment Table

This is the sample table for the CourseEnrollment Table which includes entities, attributes, relations, and descriptions.

Entity	Attribute	Data Type	Nullable	Relation to	Relation Type	Description
Course Enrollment	EnrollmentID	Primary Key	No			Unique identifier for a course enrollment.
	LearnerID	Foreign Key	Yes	Learner	One-to-Many	Learner enrolled in the course.
	CourseID	Foreign Key	Yes	Course	One-to-Many	Course in which the learner is enrolled.
	Enrollment Date	Date	Yes			Date when the enrollment occurred.
	Status	Text	Yes			Status of the enrollment (e.g., Enrolled, Completed).

Table 4.14: CourseMaterial Table

This is the sample table for the CourseMaterial Table which includes entities, attributes, relations, and descriptions.

Entity	Attribute	Data Type	Nullable	Relation to	Relation Type	Description
Course Material	MaterialID	Primary Key	No			Unique identifier for a course material.
	CourseID	Foreign Key	Yes	Course	One-to-Many	Course to which the material belongs.
	Title	Text	Yes			Title of the course material.
	Description	Text	Yes			Description of the material.
	FileURL	Text	Yes			URL or file path of the material.
	UploadDate	Date	Yes			Date when the material was uploaded.

Table 4.15: CourseWorkGraded Table

This is the sample table for the CourseWorkGraded Table which includes entities, attributes, relations, and descriptions.

Entity	Attribute	Data Type	Nullable	Relation to	Relation Type	Description
Course Work Graded	Work GradedID	Primary Key	No			Unique identifier for a graded course work.
	LearnerID	Foreign Key	Yes	Learner	One-to-Many	Learner who submitted the work.
	CourseID	Foreign Key	Yes	Course	One-to-Many	Course for which the work was submitted.
	WorkType	Text	Yes			Type of graded work (e.g., Assignment, Quiz).
	WorkTitle	Text	Yes			Title of the graded work.
	Submission Date	Date	Yes			Date when the work was submitted.
	Grade	Text	Yes			Grade received for the work.

Table 4.16: Certificate Table

This is the sample table for the Certificate Table which includes entities, attributes, relations, and descriptions.

Entity	Attribute	Data Type	Nullable	Relation to	Relation Type	Description
Certificate	CertificateID	Primary Key	No			Unique identifier for a certificate.
	LearnerID	Foreign Key	Yes	Learner	One-to-Many	Learner who earned the certificate.
	CourseID	Foreign Key	Yes	Course	One-to-Many	Course for which the certificate was earned.
	IssueDate	Date	Yes			Date when the certificate was issued.
	ExpiryDate	Date	Yes			Date when the certificate expires.

Table 4.17: CourseMeeting Table

This is the sample table for the CourseMeeting Table which includes entities, attributes, relations, and descriptions.

Entity	Attribute	Data Type	Nullable	Relation to	Relation Type	Description
Course Meeting	MeetingID	Primary Key	No			Unique identifier for a course meeting.
	CourseID	Foreign Key	Yes	Course	One-to-Many	Course for which the meeting is scheduled.
	Date	Date	Yes			Date of the meeting.
	Time	Text	Yes			Time of the meeting.
	Location	Text	Yes			Location of the meeting.

4.10 Risk Analysis

Risk analysis is a pivotal aspect of EduVerse's project planning, encompassing various potential challenges that may emerge throughout its development:

4.10.1 Technical Risks

Technical hurdles, such as technology compatibility and scalability issues, are concerns. To address compatibility risks, EduVerse must remain adaptable to evolving web standards, and for scalability, it should undergo rigorous load testing while ensuring a scalable infrastructure.

4.10.2 Business Risks

Market saturation in the competitive e-learning sector is a business risk. EduVerse must establish a unique value proposition to stand out. Additionally, devising an effective pricing strategy to attract paying users without compromising its free-access model is crucial.

4.10.3 Content and Quality Risks

Content quality and the risk of plagiarism are paramount. EduVerse needs to maintain quality through regular content reviews and robust feedback mechanisms while proactively addressing issues related to copyright violations and plagiarism in user-generated content.

4.10.4 User Adoption and Engagement Risks

Sustaining user engagement and ensuring smooth user onboarding are pivotal. EduVerse can enhance engagement through regular updates and interactive features. It must also provide a user-friendly interface and comprehensive tutorials to facilitate seamless onboarding.

4.10.5 Competition Risks

New competitors entering the market and established platforms replicating EduVerse's features are competition-related risks. Staying innovative and maintaining a competitive edge will be crucial.

To mitigate these risks effectively, EduVerse will conduct comprehensive risk assessments at different project stages, establish robust risk response plans, and maintain adaptability and flexibility in its project management approach. Continuous monitoring and periodic review of risks will be critical throughout the project's lifecycle.

4.11 Conclusion

Chapter 4 of the Software Requirement Specifications (SRS) for EduVerse offers a detailed roadmap for the platform's development. It starts off with a list of features and functional requirements, delineating specific needs for instructors, learners, and the overall system. Quality attributes, such as performance and usability, are outlined, along with the non-functional requirements and assumptions. The chapter then explores graphical user interface design, database organization through an Entity-Relationship diagram, and a data dictionary. A thorough risk analysis identifies potential challenges, fostering proactive mitigation strategies. In essence, Chapter 4 serves as a crucial guide, setting the groundwork for EduVerse's development by addressing features, requirements, design elements, and potential risks.

Chapter 5 High-Level and Low-Level Design

The complete details of high-level and low-level design of the system we are building is provided in this chapter.

5.1 System Overview

Eduverse is a comprehensive educational platform designed to revolutionize the learning experience for students, educators, and institutions. This software system aims to provide a dynamic and interactive environment that supports various aspects of the education ecosystem.

5.1.1 Learning Management System (LMS)

Eduverse incorporates a robust Learning Management System to facilitate the creation, management, and delivery of educational content. Users, including teachers and students, can access courses, assignments, quizzes, and resources in a centralized and user-friendly interface.

5.1.2 Collaborative Learning Tools

To enhance the collaborative learning experience, Eduverse integrates tools such as discussion forums, group projects, and real-time collaboration features. This encourages student engagement and fosters a sense of community within the platform.

5.1.3 Personalized Learning Paths

The system employs intelligent algorithms to analyze user behavior and preferences, offering personalized learning paths and content recommendations. This adaptive approach ensures that each user receives a tailored educational experience.

5.1.4 Assessment and Feedback

Eduverse supports various assessment methods, including quizzes, exams, and assignments. Additionally, it provides instant feedback to students and detailed analytics to educators, enabling data-driven decision-making to enhance the learning process.

5.2 Design Considerations

This section outlines important considerations that need to be addressed during the design phase of Eduverse.

5.2.1 Assumptions and Dependencies

The design of Eduverse relies on several assumptions and dependencies to ensure its proper functioning:

- **Related Software or Hardware:** Eduverse assumes compatibility with modern browsers and devices commonly used for educational purposes.
- **Operating Systems:** The system is designed to operate seamlessly on major operating systems, including Windows, macOS, and Linux.
- **End-user Characteristics:** Eduverse assumes that end-users have basic computer literacy and internet access.
- **Possible and/or Probable Changes in Functionality:** The design considers the possibility of future updates and expansions to accommodate evolving educational needs.

5.2.2 General Constraints

Eduverse operates within certain constraints that influence its design and functionality:

- **Hardware or software environment:**The system is designed to work within the capabilities of standard educational hardware and software.
- **End-user environment:** Eduverse is optimized for diverse end-user environments, considering variations in network conditions and device specifications.
- **Standards compliance:** The design adheres to relevant educational standards and specifications to ensure interoperability.
- **Security requirements:** Eduverse incorporates robust security measures to protect user data and maintain compliance with privacy regulations.
- **Performance requirements:** The system is optimized for efficient performance to provide a seamless learning experience.

5.2.3 Goals and Guidelines

Eduverse is designed with several key goals, guidelines, principles, and priorities to ensure its effectiveness and user satisfaction.

- **The KISS principle (”Keep it simple stupid!”):** Simplicity is crucial for user adoption and ease of use. A simple and intuitive interface ensures that users can quickly navigate the platform without unnecessary complexity.

- **Emphasis on speed versus memory use:** Eduverse prioritizes performance to provide a seamless user experience. Fast response times and efficient memory usage contribute to a responsive and reliable platform.
- **Working, looking, or ”feeling” like an existing product:** Familiarity promotes user comfort. Designing Eduverse with a look and feel similar to widely used educational platforms enhances user adoption and reduces the learning curve.

5.2.4 Development Methods

The development of Eduverse follows a systematic and well-defined approach to ensure robustness and scalability.

- **Agile Development Methodology:** Eduverse employs the Agile methodology to adapt to changing requirements, promote collaboration, and deliver incremental updates. This approach allows for continuous improvement based on user feedback.
- **Iterative Design Process:** Eduverse undergoes iterative design to refine features and functionalities. Each iteration incorporates user feedback, ensuring that the platform evolves to meet the dynamic needs of the education sector.
- **User-Centered Design (UCD):** UCD is integral to Eduverse’s development, focusing on understanding user behaviors, needs, and preferences. This approach ensures that the platform aligns with the expectations and requirements of its users.

5.3 System Architecture

The system Architecture of EduVerse is given which highlights how the system will process data and interact with components within the system.

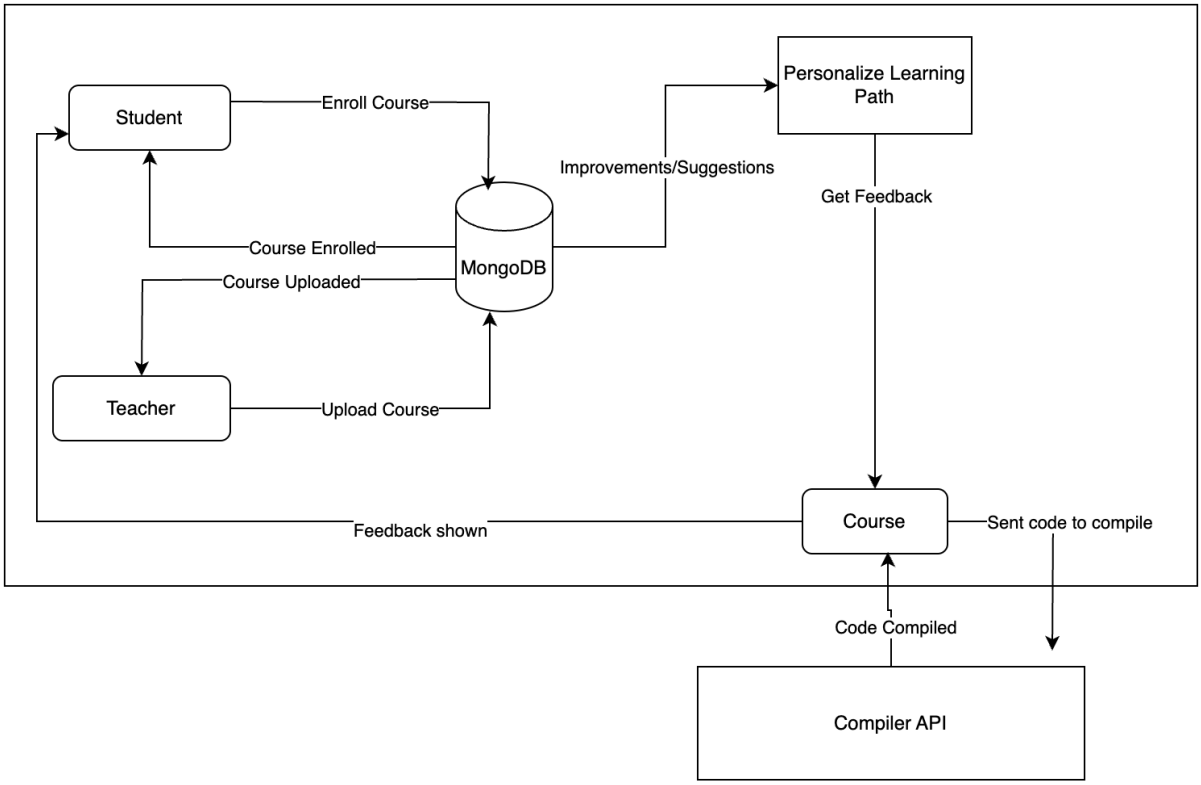


Figure 5.1: Architecture Diagram of EduVerse Platform
Comprehensive visualization of the architectural components and their interactions within EduVerse, providing an overview of the system's design and functionality.

The system architecture of Eduverse is designed to provide a robust and scalable platform for delivering educational services. The architecture encompasses both internal and external aspects, focusing on the effective partitioning of functionality into subsystems and components.

Top-Level Responsibilities

At the top-most level, Eduverse is responsible for the following major functionalities:

- **User Management:** Handles user registration, authentication, and authorization.
- **Content Delivery:** Manages the storage, retrieval, and distribution of educational content.
- **Collaboration Tools:** Facilitates real-time collaboration between students and educators.

Rationale

The decomposition is driven by the need for modularity and scalability. Each subsystem addresses a specific aspect of the educational experience, allowing for independent development and scaling. The

chosen architecture promotes flexibility in accommodating future enhancements and aligns with the goals of simplicity and performance.

5.3.1 Subsystem Architecture

- **Authentication Subsystem** The Authentication Subsystem is a critical component ensuring secure user access to Eduverse
  - **Internal Components:**
    * User Credentials Management
    * Authentication Services
    * Access Control
  - **Interactions:**
    * User submits credentials.
    * Authentication service validates credentials.
    * Access control determines user permissions.
- **Content Delivery Subsystem** Responsible for the storage, retrieval, and delivery of educational content.
  - **Internal Components:**
    * Content Storage
    * Retrieval Services
    * Distribution Mechanism
  - **Interactions:**
    * Content is uploaded and stored
    * User requests content
    * Distribution mechanism ensures efficient delivery

5.4 Architectural Strategies

The architectural strategies that will be used to access EduVerse are mentioned below:

5.4.1 Tools and Software

Policy/Tactic-1: MERN Stack Implementation

Description: MERN stack will be used to build the system as it is widely used in the software field for developing websites. MERN is based on component-based programming. MongoDB will be employed for data access in a NO-SQL fashion. VScode will be used as a code editor due to its provision of different extensions, features, and compatibility with database and API connections.

Requirements: Internet connection is necessary to access the database.

5.4.2 Hardware

Policy/Tactic-2: System Requirements

Description: A personal computer (PC) or a laptop with an internet connection is necessary to use the EduVerse product. The hardware should also support an internet browser.

5.4.3 Future Plans

Policy/Tactic-3: Expansion and AI Integration

Description: In the future, the product will be scaled to cover multiple courses around different programming languages, such as Python and C. Additionally, features utilizing AI will be introduced to automate various aspects of the platform.

These architectural strategies outline the tools, software, and hardware requirements for accessing EduVerse, along with future plans for expansion and integration of AI-driven features.

5.5 Domain Model/Class Diagram

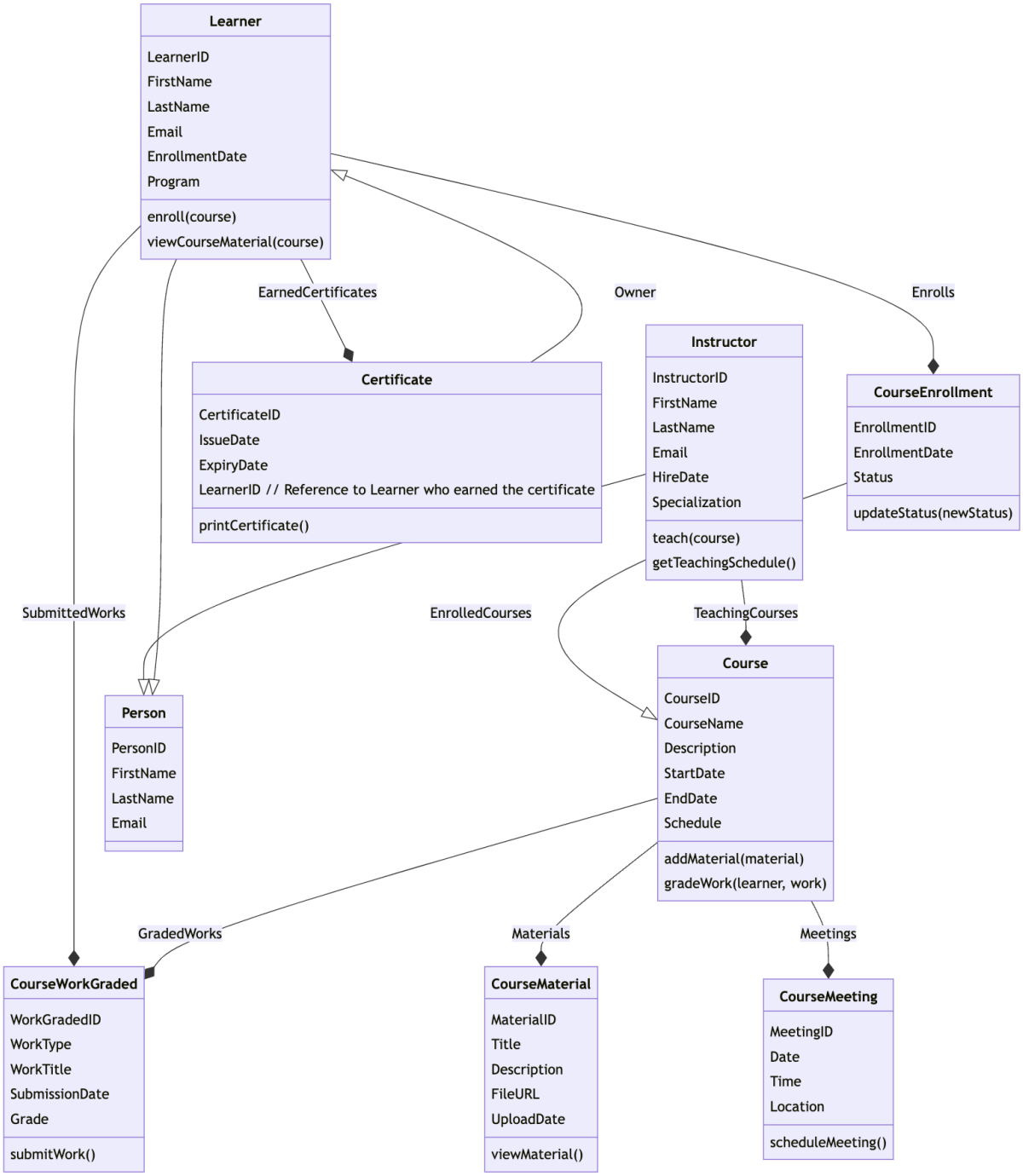


Figure 5.2: Class Diagram

The class diagram of whole system is given above

5.6 Sequence Diagrams

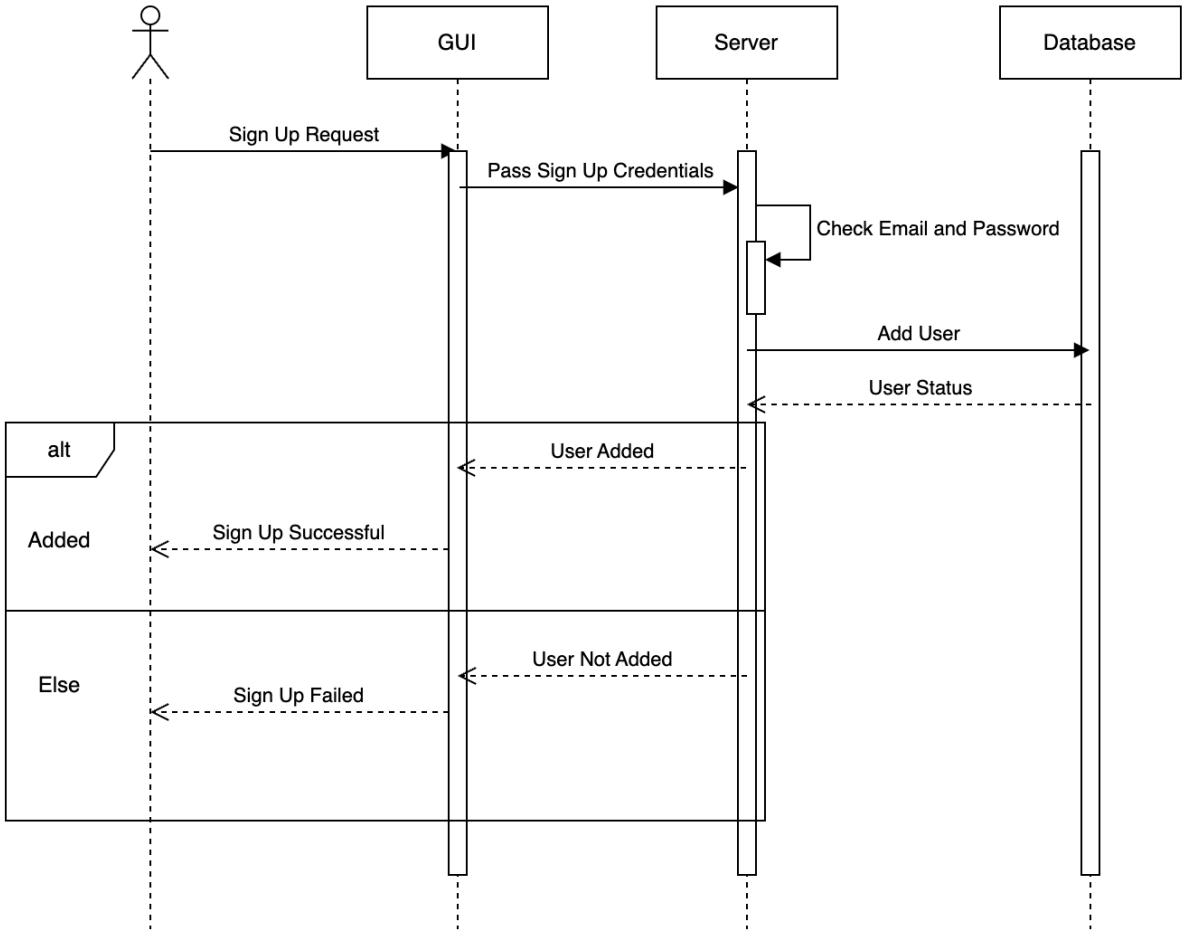


Figure 5.3: Sequence Diagram for User Registration

Illustration of the step-by-step process involved in user registration on EduVerse, highlighting interactions between system components during the sign-up procedure.

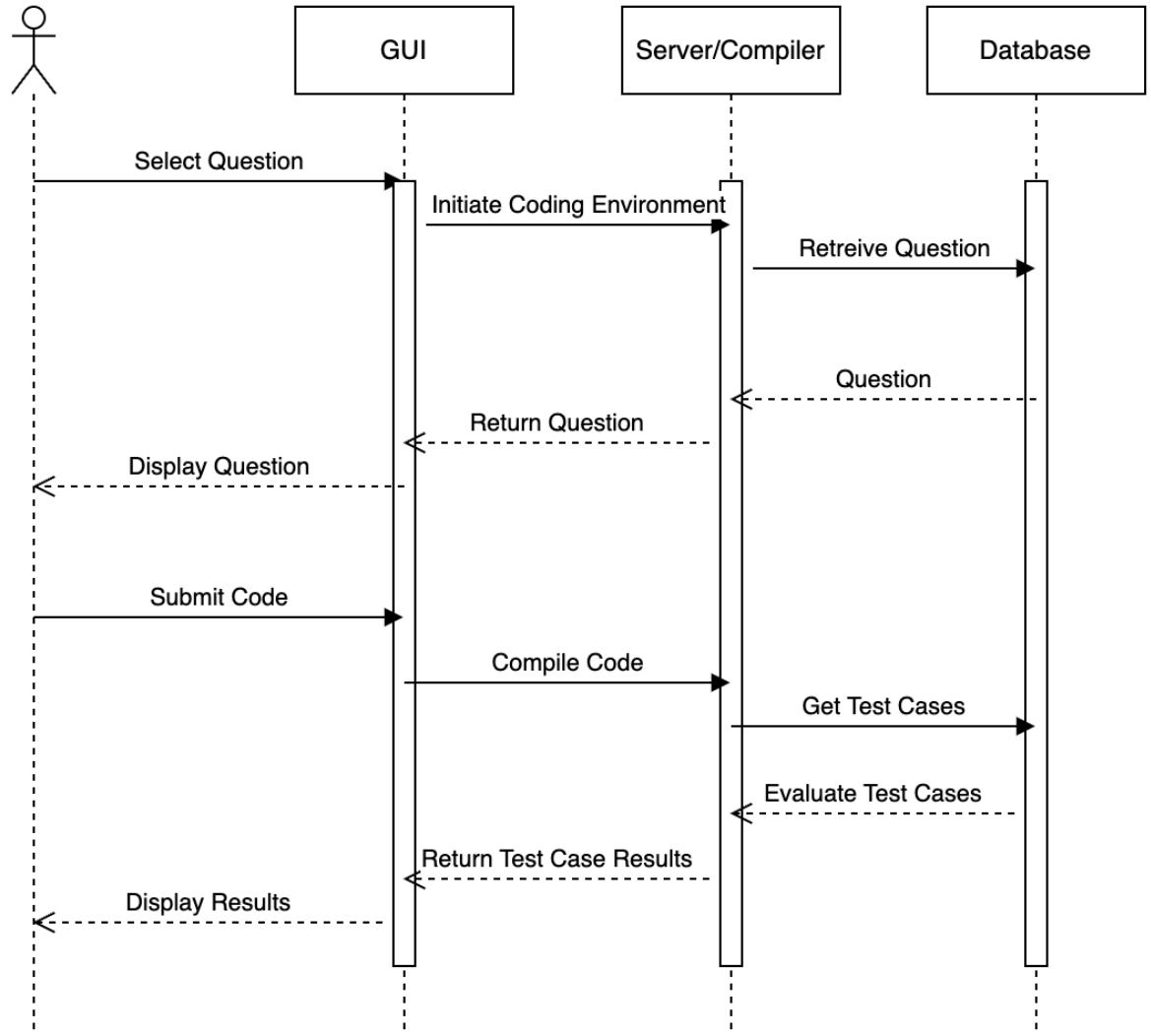


Figure 5.4: Sequence Diagram for Coding Editor Interaction

Visual representation of the dynamic interactions within EduVerse's coding editor, showcasing the sequence of events.

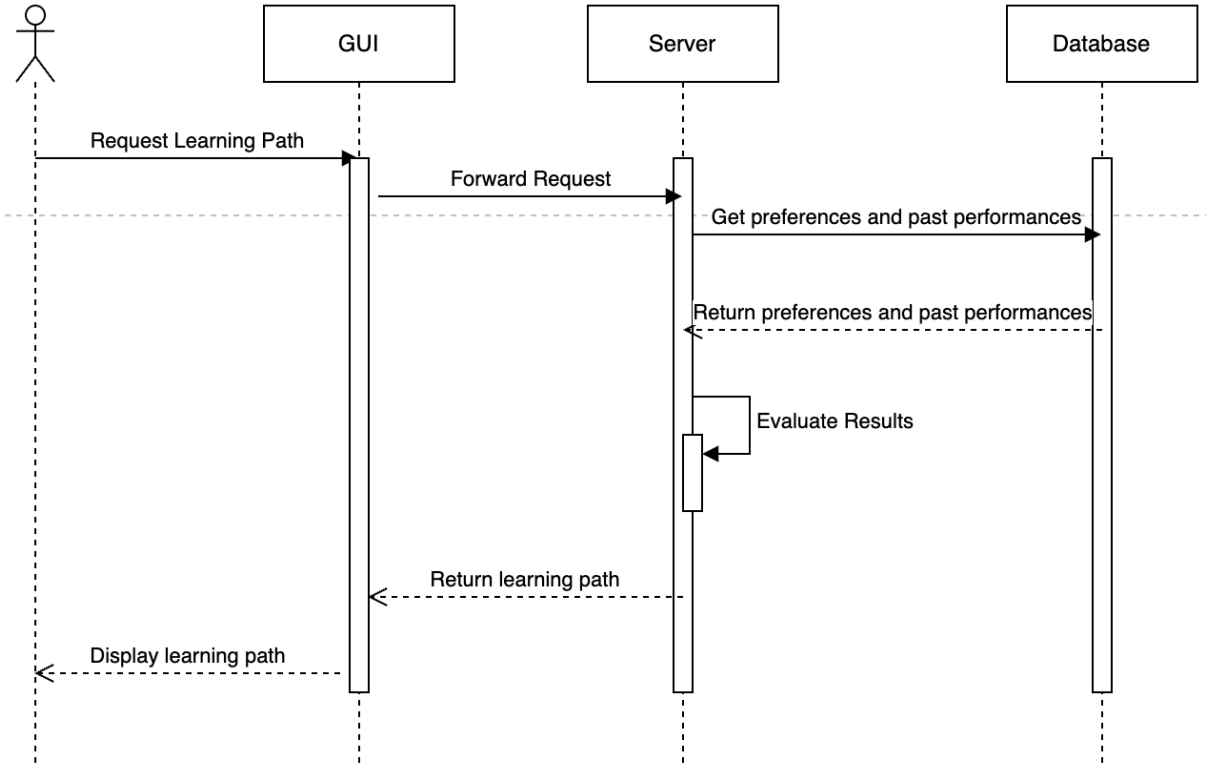


Figure 5.5: Sequence Diagram for Learning Path Progress
Illustration of the dynamic interactions within EduVerse's learning path module.

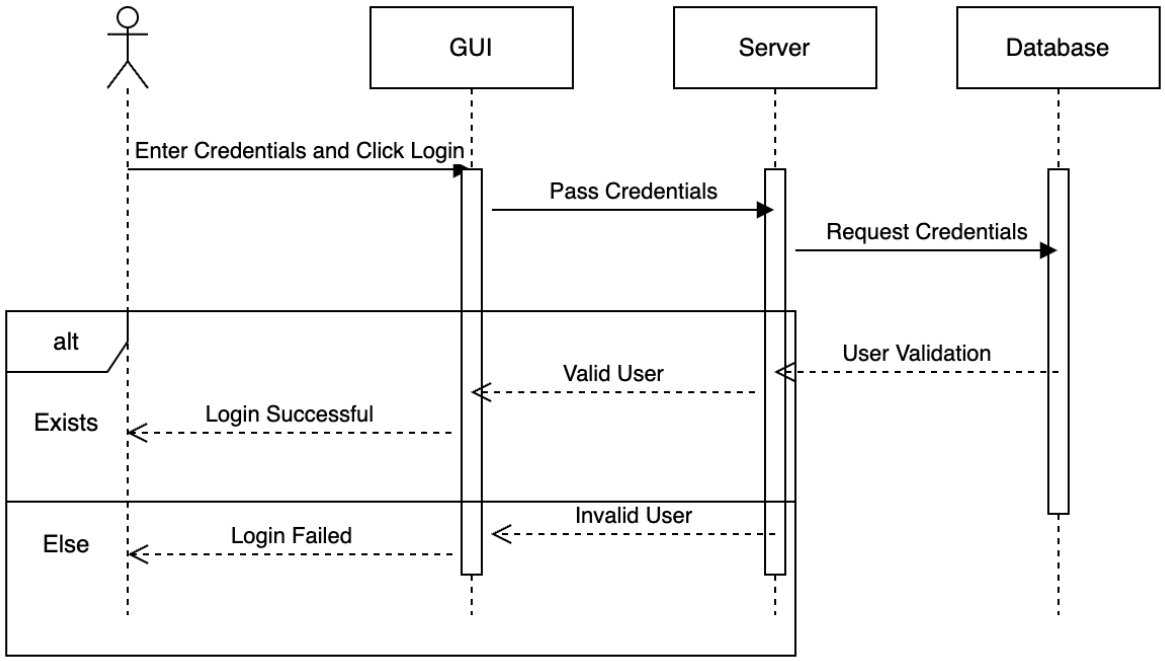


Figure 5.6: Sequence Diagram for User Login
Visual representation of the dynamic interactions within EduVerse's login process, showcasing the sequence of events as users authenticate and access the platform's features.

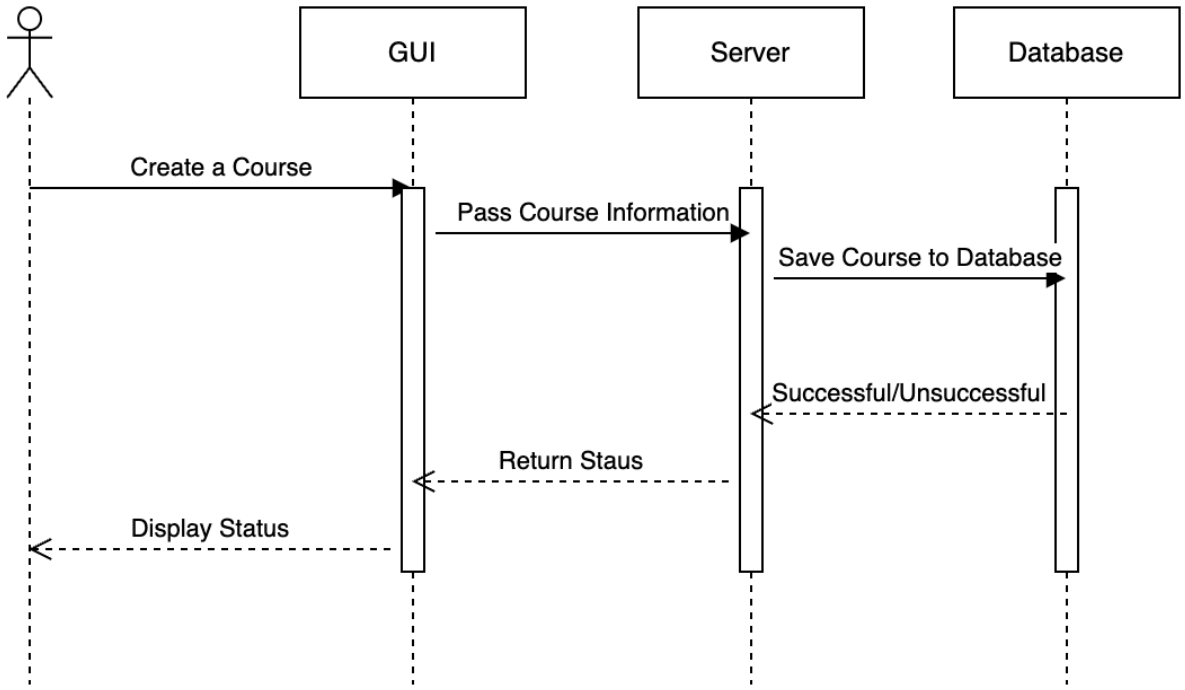


Figure 5.7: Sequence Diagram for Course Creation

Illustration of the dynamic interactions within EduVerse's course creation process, highlighting the sequence of events involved in setting up a new educational course.

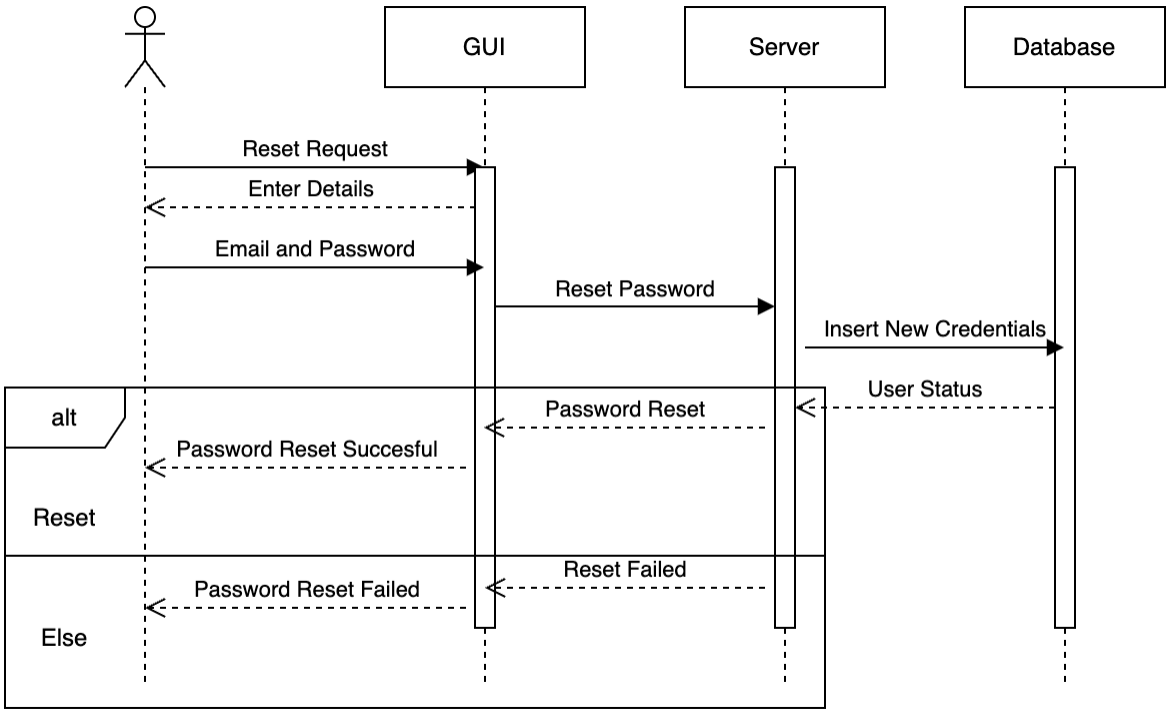


Figure 5.8: Sequence Diagram for Password Reset

Visual representation of the dynamic interactions within EduVerse's password reset process, depicting the sequence of events as users initiate and complete the password recovery procedure.

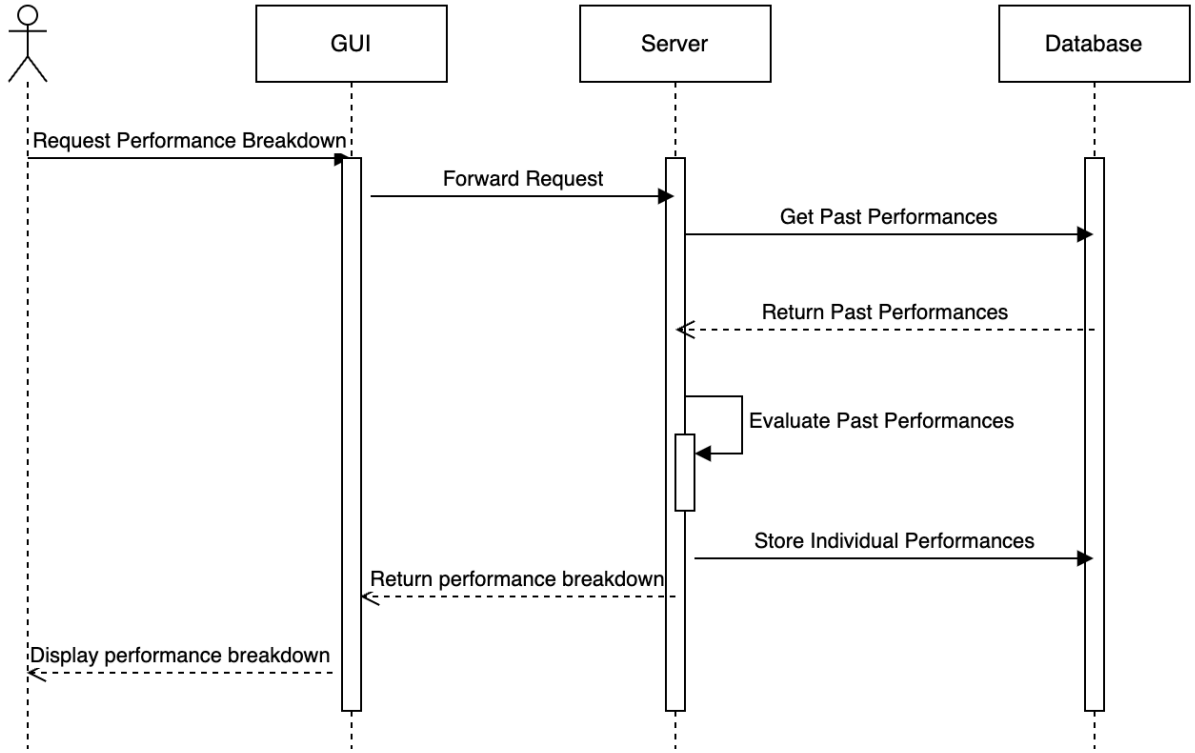


Figure 5.9: Sequence Diagram for Performance Breakdown Analysis

Illustration of the dynamic interactions within EduVerse's performance breakdown analysis, highlighting the sequence of events involved in assessing and analyzing system performance metrics.

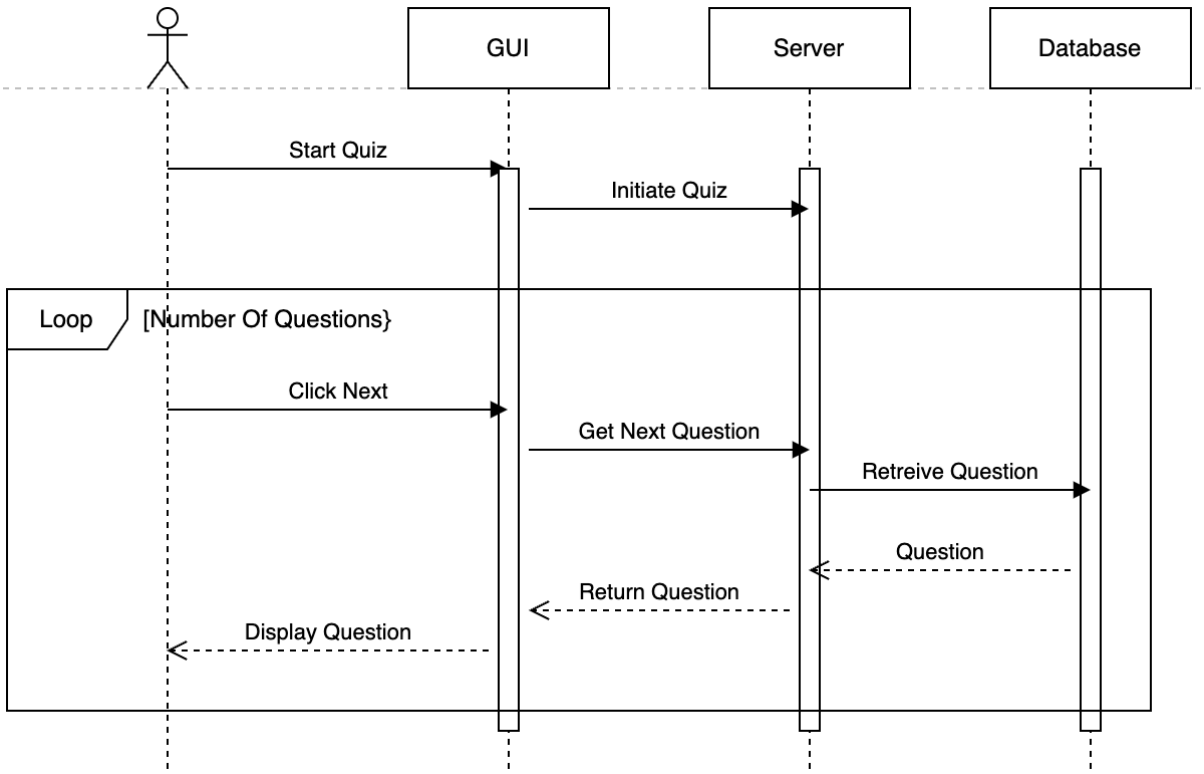


Figure 5.10: Sequence Diagram for Quiz Interaction

Visual representation of the dynamic interactions within EduVerse's quiz process, showcasing the sequence of events as users engage in quizzes and submit answers.

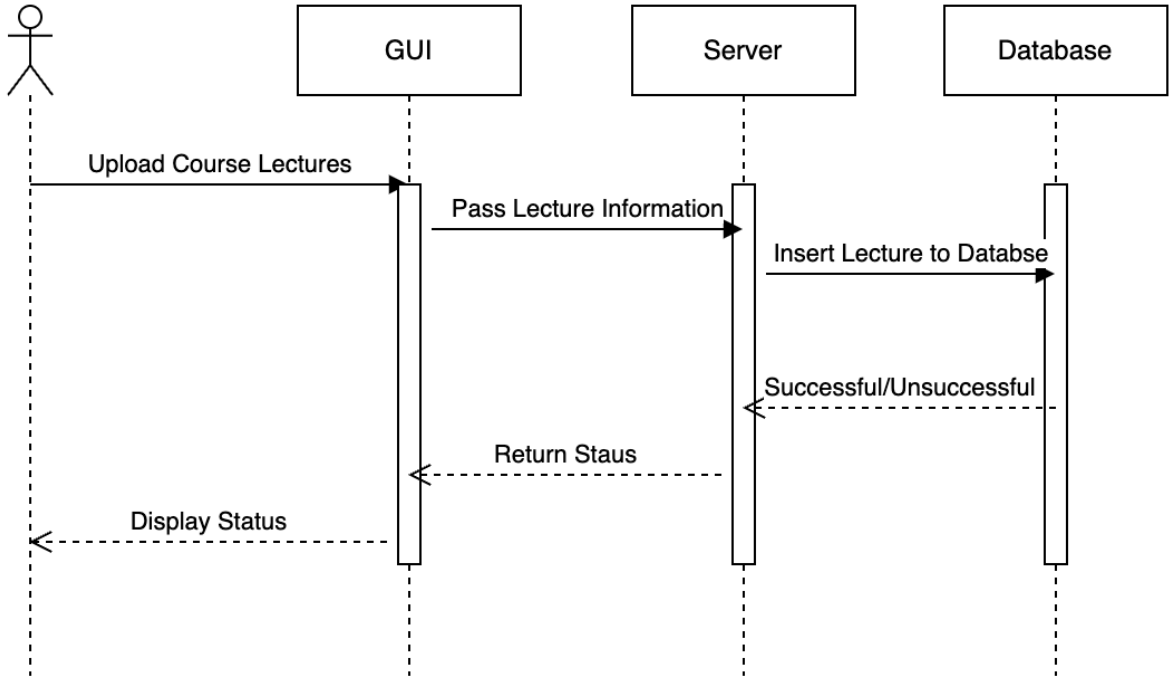


Figure 5.11: Sequence Diagram for Lecture Upload Process

Illustration of the dynamic interactions within EduVerse's lecture upload process, highlighting the sequence of events involved as users contribute educational content to the platform.

5.7 Policies and Tactics

This section outlines specific design policies and tactics that influence the detailed aspects of the Eduverse system without significantly impacting its overall architecture. These decisions encompass various elements, including the choice of tools, engineering trade-offs, coding conventions, and plans for testing and maintenance.

5.7.1 Codebase Organization

Policy/Tactic-1: Hierarchical Source Code Organization

Description: The source code of Eduverse is organized in a hierarchical structure, with clear divisions into logical components, files, and directories. This organization enhances readability, maintainability, and collaborative development.

Reasoning: A well-structured codebase simplifies navigation, encourages modular development, and facilitates teamwork. This policy ensures that developers can easily locate and comprehend relevant sections of the code, leading to more efficient development and troubleshooting.

5.7.2 Development Tools

Policy/Tactic-2: Standardized Development Tools

Description: Eduverse adopts standardized development tools, including compilers, interpreters, and libraries. These tools are carefully selected to ensure compatibility, reliability, and optimal performance throughout the development life cycle.

Reasoning: Standardizing development tools promotes consistency, streamlines the development process, and reduces the likelihood of compatibility issues. By using established and widely-accepted tools, Eduverse ensures a stable and well-supported development environment.

5.7.3 Coding Guidelines

Policy/Tactic-3: Enforced Coding Guidelines

Description: Eduverse enforces coding guidelines and conventions to maintain a uniform coding style across the entire codebase. These guidelines cover aspects such as naming conventions, indentation, and commenting.

Reasoning: Consistent coding practices enhance code readability, ease collaboration among developers, and contribute to the overall quality of the software. Enforcing coding guidelines is essential for creating

a maintainable and comprehensible codebase.

5.7.4 Testing Strategies

Policy/Tactic-4: Comprehensive Testing Plans

Description: Eduverse incorporates comprehensive testing plans that include unit testing, integration testing, and system testing. Automated testing tools are employed to ensure the reliability and functionality of the system.

Reasoning: Rigorous testing is crucial to identify and rectify software defects promptly. Automated testing streamlines the testing process, enabling faster feedback to developers and facilitating continuous integration practices.

5.7.5 Documentation Practices

Policy/Tactic-5: Thorough Documentation Practices

Description: Eduverse prioritizes thorough documentation, including inline comments, API documentation, and user guides. This documentation ensures that developers, administrators, and end-users have access to clear and comprehensive information.

Reasoning: Effective documentation facilitates understanding, maintenance, and collaboration. It serves as a valuable resource for both the development team and end-users, contributing to the long-term success of the Eduverse platform.

These policies and tactics collectively contribute to the detailed design and implementation of Eduverse, fostering a development environment that prioritizes clarity, consistency, and reliability.

5.8 Conclusion

Chapter 5 presents a comprehensive exploration of the high-level and low-level design aspects for EduVerse. The system overview outlines key components, including the Learning Management System (LMS), personalized learning paths, and assessment and feedback mechanisms. Design considerations, encompassing assumptions, dependencies, constraints, goals, and guidelines, provide a robust foundation. The system architecture and subsystem architecture are detailed, highlighting the structural organization. Architectural strategies, involving tools, software, hardware, and future plans, shape the development approach. The domain model/class diagram and sequence diagrams offer visual representations of the system's structure and interactions. Policies and tactics, addressing codebase organization, development tools, coding guidelines, testing strategies, and documentation practices, ensure a system-

atic development process. In conclusion, Chapter 5 lays the groundwork for the design phase, providing a detailed blueprint for EduVerse's architecture, development, and implementation.

Chapter 6 Implementation and Test Cases

The implementation details of the website created and interaction between the modules as well as the coding environment designed is described here in the implementation section 6.1.

6.1 Implementation

The implementation details of the components and algorithms of prototype designed is given in subsection 6.1.1.

6.1.1 Prototype Description

The prototype for EduVerse developed in Final Year Project 2 (FYP-2) successfully delivers a complete frontend experience for both teachers and students. The backend infrastructure ensures seamless user interactions, featuring essential functionalities such as user account creation, course management, and storage of educational materials. Additionally, a Code Editor with an in-browser C++ compiler enhances the platform, laying the foundation for robust coding experiences in subsequent development phases.

6.1.2 LMS Module

This section outlines the integration of frontend and backend components within EduVerse's Learning Management System (LMS). The frontend design ensures an intuitive and user-friendly interface, incorporating features for live session scheduling, uploading of course materials, streamlined student feedback mechanisms, and automated certificate issuance. On the backend, functionalities ranging from live session management to secure storage and retrieval of course materials, fostering an interactive and comprehensive educational environment.

6.1.3 Code Editor Module

This module delineates the architecture and functionalities of EduVerse's embedded Code Editor. Offering an in-browser C++ compiler, this feature is pivotal for coding exercises and assignments.

6.2 Test case Design and description

All the test cases performed during the testing phase of the system are elaborated in the sub chapters below.

Table 6.1: Test case No.1: Registration

This table provides a test case for the registration flow in the website's login module. It outlines the steps for user registration and verifies that the details are stored in the database.

Registration Module			
Test Case ID:	*1*	**QA Test Engineer:**	*Abdullah Waseem*
Test case Version:	*1*	**Reviewed By:**	*Aroobha Amjad*
Test Date:	*30-04-2024*	**Use Case Reference(s):**	*4.7.1 - User Registration*
Revision History:	*None*		
Objective:	*Test the Registration Flow*		
Product/Ver/ Module:	*Website Login Module*		
Environment:	*Website is running on any web browser and internet is connected.*		
Assumptions:	*User is connected to the internet*		
Pre-Requisite:	*No pre-requisite required*		
Step No.	**Execution description**	**Procedure result**	
1.	*User enters their details such as email, name, password*	*Details stored in database and user redirected to home page*	
Comments:			
☐ **Passed** ☐ **Failed** ☐ **Not Executed**			

Table 6.2: Test case No.2: Login

This table presents a test case for the login flow in the website's login module. It describes the steps for user login and verifies that the system redirects the user to the home screen upon successful login.

Login, Logout Module			
Test Case ID:	*2*	**QA Test Engineer:**	*Fahad Waseem*
Test case Version:	*1*	**Reviewed By:**	*Aroobha Amjad*
Test Date:	*30-04-2024*	**Use Case Reference(s):**	*4.7.2 - User Login*
Revision History:	*None*		
Objective:	*Test the Login Flow*		
Product/Ver/ Module:	*Website Login Module*		
Environment:	*Website is running on any web browser and internet is connected.*		
Assumptions:	*User is connected to the internet*		
Pre-Requisite:	*User has already registered*		
Step No.	**Execution description**	**Procedure result**	
1.	*User enters their credentials, i.e., their username and password, and clicks login.*	*System takes the user to the home screen.*	
Comments:			
☐ **Passed** ☐ **Failed** ☐ **Not Executed**			

Table 6.3: Test case No.3: Logout

This table presents a test case for the logout flow in the website's logout module. It describes the step where the user clicks on the logout button, and verifies that the system redirects the user to the login.

Login, Logout Module			
Test Case ID:	*3*	**QA Test Engineer:**	*Fahad Waseem*
Test case Version:	*1*	**Reviewed By:**	*Aroobha Amjad*
Test Date:	*30-04-2024*	**Use Case Reference(s):**	*4.7.2 - User Login*
Revision History:	*None*		
Objective:	*Test the Logout Flow*		
Product/Ver/ Module:	*Website Logout Module*		
Environment:	*Website is running on any web browser and internet is connected.*		
Assumptions:	*User is connected to the internet and has logged in*		
Pre-Requisite:	*User has already registered and is logged in*		
Step No.	**Execution description**	**Procedure result**	
1.	*Users click on the logout button.*	*System takes the user to the login page.*	
Comments:			
■ **Passed** ☐ **Failed** ☐ **Not Executed**			

Table 6.4: Test case No.4: Course Enrollment

This table presents a test case for the course enrollment functionality in the website's course enrollment module.

Enroll in a Course			
Test Case ID:	*4*	**QA Test Engineer:**	*Fahad Waseem*
Test case Version:	*1*	**Reviewed By:**	*Aroobha Amjad*
Test Date:	*30-04-2024*	**Use Case Reference(s):**	*4.7.3 - Course Enrollment Request*
Revision History:	*None*		
Objective:	*Test the Course Enrollment Functionality*		
Product/Ver/ Module:	*Course Enrollment Module*		
Environment:	*Website is running on any web browser and internet is connected.*		
Assumptions:	*User is connected as a student*		
Pre-Requisite:	*User has already registered and is logged in*		
Step No.	**Execution description**	**Procedure result**	
1.	*Users clicks on the enroll button*	*System enrolls the user into the course.*	
Comments:			
■ **Passed** ☐ **Failed** ☐ **Not Executed**			

Table 6.5: Test case No.5: Course Creation

This table presents a test case for the course creation functionality in the website's course creation module.

Course Creation Module			
Test Case ID:	*5*	**QA Test Engineer:**	*Adbullah Waseem*
Test case Version:	*1*	**Reviewed By:**	*Aroobha Amjad*
Test Date:	*27-04-2024*	**Use Case Reference(s):**	*4.7.4 - Creating a New Course*
Revision History:	*None*		
Objective:	*Test the Course Creation Flow*		
Product/Ver/ Module:	*Course Creation Module*		
Environment:	*Website is running on any web browser and internet is connected.*		
Assumptions:	*User is signed up as a teacher*		
Pre-Requisite:	*User has already registered*		
Step No.	**Execution description**	**Procedure result**	
1.	*The user will fill out all the required information required for the course.*	*A new job post will be created successfully.*	
Comments:			
☑ **Passed** ☐ **Failed** ☐ **Not Executed**			

Table 6.6: Test case No.6: Meet Scheduling

This table presents a test case for the live meeting scheduling functionality in the website's live meeting scheduling module. It describes the step where the instructor fills out the required details for the live session, and verifies that the live meeting is scheduled and accessible/viewable by students.

Live Meeting Scheduling Module			
Test Case ID:	*6*	**QA Test Engineer:**	*Fahad Waseem*
Test case Version:	*1*	**Reviewed By:**	*Aroobha Amjad*
Test Date:	*29-04-2024*	**Use Case Reference(s):**	*4.7.5 - Scheduling a Live Session*
Revision History:	*None*		
Objective:	*Test the Live Meeting Flow*		
Product/Ver/ Module:	*Course Content Uploading Module*		
Environment:	*Website is running on any web browser and internet is connected.*		
Assumptions:	*User is signed up as a teacher*		
Pre-Requisite:	*User has already registered*		
Step No.	**Execution description**	**Procedure result**	
1.	*The instructor fills out the required details for the Meet Live Session.*	*Live Meeting will be scheduled and students can access/view it.*	
Comments:			
☑ **Passed** ☐ **Failed** ☐ **Not Executed**			

Table 6.7: Test case No.7: Material Uploading

This table presents a test case for the course content uploading functionality in the website's course content uploading module. It describes the step where the instructor uploads the course content, and verifies that the course content is stored in the database.

Course Content Uploading Module			
Test Case ID:	*7*	**QA Test Engineer:**	*Fahad Waseem*
Test case Version:	*1*	**Reviewed By:**	*Aroobha Amjad*
Test Date:	*30-04-2024*	**Use Case Reference(s):**	*4.7.6 - Uploading Course Materials*
Revision History:	*None*		
Objective:	*Test the Material Uploading Flow*		
Product/Ver/ Module:	*Course Content Uploading Module*		
Environment:	*Website is running on any web browser and internet is connected.*		
Assumptions:	*User is signed up as a teacher*		
Pre-Requisite:	*User has already registered*		
Step No.	**Execution description**	**Procedure result**	
1.	*The instructor uploads the course content.*	*Course content will be stored in the database.*	
Comments:			
	☑ **Passed** ☐ **Failed** ☐ **Not Executed**		

Table 6.8: Test case No.8: Profile

This table presents a test case for the user profile module in the website. It describes the step where the user updates their details, such as their password, and verifies that the changes are reflected in the database as well as on the profile screen.

User Profile Module			
Test Case ID:	*8*	**QA Test Engineer:**	*Abdullah Waseem*
Test case Version:	*1*	**Reviewed By:**	*Aroobha Amjad*
Test Date:	*29-04-2024*	**Use Case Reference(s):**	*4.7.7 - Managing Account Settings*
Revision History:	*None*		
Objective:	*Test the User Info Updation Flow*		
Product/Ver/ Module:	*User Info Module*		
Environment:	*Website is running on any web browser and internet is connected.*		
Assumptions:	*User is signed up*		
Pre-Requisite:	*User has already logged in and is on the Profile Page*		
Step No.	**Execution description**	**Procedure result**	
1.	*User Updates their details, such as their password.*	*Changes will be reflected on the database as well as the profile screen.*	
Comments:			
	☑ **Passed** ☐ **Failed** ☐ **Not Executed**		

Table 6.9: Test case No.9: Certificates

This table presents a test case for the certificates module in the website.

Certificates Module			
Test Case ID:	*9*	**QA Test Engineer:**	*Fahad Waseem*
Test case Version:	*1*	**Reviewed By:**	*Aroobha Amjad*
Test Date:	*30-04-2024*	**Use Case Reference(s):**	*4.7.8 - Certificates*
Revision History:	*None*		
Objective:	*Test the Certificates Creation and Viewing*		
Product/Ver/ Module:	*Certificates Module*		
Environment:	*Website is running on any web browser and internet is connected.*		
Assumptions:	*User is signed up*		
Pre-Requisite:	*User has completed a course.*		
Step No.	**Execution description**	**Procedure result**	
1.	*User Completes course and generates Certificate*	*Certificate will be made available.*	
Comments:			
☑ **Passed** ☐ **Failed** ☐ **Not Executed**			

Table 6.10: Test case No.9: Quiz

This table presents a test case for the Quiz module in the website. Quizzes can be created by instructors and attempted by learners.

Certificates Module			
Test Case ID:	*9*	**QA Test Engineer:**	*Abdullah Waseem*
Test case Version:	*1*	**Reviewed By:**	*Aroobha Amjad*
Test Date:	*30-04-2024*	**Use Case Reference(s):**	*4.7.9 - Quiz*
Revision History:	*None*		
Objective:	*Test the Quiz Creation and Attempting*		
Product/Ver/ Module:	*Quiz Module*		
Environment:	*Website is running on any web browser and internet is connected.*		
Assumptions:	*User is signed up*		
Pre-Requisite:	*User has access to a course.*		
Step No.	**Execution description**	**Procedure result**	
1.	*User attempts a Quiz.*	*Timer and Quiz checking is done effectively and scores are displayed.*	
2.	*User creates a Quiz.*	*User has the ability to set up questions or upload a document with the questions.*	
Comments:			
☑ **Passed** ☐ **Failed** ☐ **Not Executed**			

6.3 Test Metrics

Summarize here the common ground of attributes of test case metrics as explained in Table 6.11.

Table 6.11: Test case Matric

This Test Case Matric provides information of the test cases and their scores.

Metric	**Value**
Number of Test Cases	9
Number of Test Cases Passed	9
Number of Test Cases Failed	0
Test Case Defect Density	0
Test Case Effectiveness	0
Traceability Matrix	File is enclosed separately.

Chapter 7 User Manual

7.1 Sign-Up

1. On the website, look for the Signup button. This is located in the upper-right corner of the page.
2. To open the signup form, click the Signup button.
3. Fill out the registration form with all required information, such as your name, email address, role, and password. Ensure that your password is strong, with a mix of capital and lowercase letters, numbers, and symbols.
4. Once all the necessary fields have been filled out, click ”Signup” to begin the account creation process.
5. Your browser will redirect you to the form for better recommendation process. You can either fill it or skip it.
6. Then you will be redirected to the home page.

7.2 Login

1. Find the Login button on the website by visiting it. This is found in the page’s upper right corner.
2. To open the login form, click the Login button.
3. Enter your password and email address in the designated fields. Click the ”Forgot password” link to reset your password if you’ve forgotten it.
4. After providing your login details, click the ”Login” button to gain access to your account.
5. After clicking on Login button, you will be redirected to your homepage

7.3 Course Enrollment

1. You login to your account by providing you email and password.
2. Then you click on the ”Courses” section from the navbar located on top of the page.
3. Here you can see all the courses you are already enrolled in and all the courses you can enroll in.
4. You click the ”Read more” button on the course you want to be enrolled in.
5. You will be taken to course details page where you can click the ”Enroll” button.

7.4 Course Creation

1. You login to your account by providing you email and password.
2. Then you click on the ”Courses” section from the navbar located on top of the page.
3. Here you can see all the courses you have already created, and below that you have the button to create a new course.
4. You click the ”Create New Course” button.
5. You will be taken to a form where you provide course name, affilizated university and course length.
6. After filling the form you click on the ”Create Course” button.
7. You will be redirected to course homepage where you can upload material, quizzez and comments for you students to see.

7.5 Create Live Session

1. You login to your account by providing you email and password.
2. Then you click on the ”Courses” section from the navbar located on top of the page.
3. Here you can see all the courses you have already created.
4. Click on the course you want to create a live session in.
5. You will be redirected to the homepage.
6. In course home you can comment the link to your live session and students can join from there.

7.6 Upload Course Material and Quizzes

1. You login to your account by providing you email and password.
2. Then you click on the ”Courses” section from the navbar located on top of the page.
3. Here you can see all the courses you have already created.
4. Click on the course you want to upload the material in.
5. You will be redirected to the homepage.
6. On left side you can see a sidebar, you have to select ”Materials” to upload material for course or

”Quizzes” to upload quizzes for your course.

7. In Materials you can see week wise sections and a button to upload the file.
8. After uploading the file, click on the ”Upload” button.
9. If you want to upload more than one material for a week then you can click on ”Add New Materials” and repeat the last step.
10. In Quizzes section, you can see already made quizzes and a button ”Add New Quiz”
11. After clicking on this button, you have to enter Quiz name, time and upload a file with questions.
12. Then you simply click on ”Save Quiz” button.

7.7 Manage Account Settings

1. You login to your account by providing you email and password.
2. On top right corner, you can see your name. Click on your name.
3. You will be redirected to your profile page.
4. Here you can select the ”Edit” button.
5. Now you can update your name, email or password.

7.8 Certificates

1. You Completes a Course.
2. You click on the Generate Quiz Button.
3. Course Completion is verified.
4. Certificate is displayed to you

7.9 Attempt Quiz

1. You login to your account by providing you email and password.
2. Then you click on the ”Courses” section from the navbar located on top of the page.
3. Here you can see all the courses you have already enrolled in.
4. Click on the course you want to attempt a quiz in.

5. You will be redirected to the homepage.
6. On left side you can see a sidebar, you have to select ”Quizzes”.
7. Here you can see all the quizzes that your instructor has created.
8. Select on the quiz name relevant to you, and you will be redirected to the quiz page where you can attempt the quiz.

Chapter 8 Conclusion and Future Work

This chapter concludes by reiterating the work done and prototype developed so far and also highlights the plans we intend to work on in the future.

8.1 Conclusion

There has been good amount of work done to automated basic recruitment system every work has its pros and cons based on its scope. A thorough analysis of existing learning platforms has been done and the strategies they used to target different groups have also been reviewed.

The software requirements for the system are also documented along with the important features of the project and other hardware and software requirements. This document includes low level and high-level design and er-diagram of system.

The development work involves creating and putting together key parts of EduVerse, a solid e-learning platform. We've built a user-friendly website for teachers and students to use easily. On the technical side, we've set up the basics, such as allowing users to create accounts, managing course details, and storing lecture materials like slides and videos. A differentiating feature is a Code Editor that lets users write and compile C++ code directly on our website. This makes learning more interactive and practical. Overall, our goal is to provide a complete and useful educational platform.

The functionalities related to adding support for different programming langauges can be extended in the future by groups that might be interested in this idea.

8.2 Future Work

Several avenues for future work present exciting opportunities for EduVerse. The focus should be on refining the code editor implementation, incorporating robust test cases and black-box testing for assignments to ensure a seamless user experience. Further development is needed for personalized learning paths, offering tailored educational journeys for individual users. Implementation of performance breakdowns, will provide learners with tangible achievements. Additionally, the integration of quiz competitions will enhance engagement and interactivity within the platform. These proposed enhancements will contribute to the continuous growth and effectiveness of EduVerse.

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